

NE 511: Multi-Physics of Nuclear Reactors

UNIT 5: Novel Multi-Physics Modeling

*Lecture 18: HIGH-FIDELITY NEUTRONICS, THERMAL-HYDRAULICS, AND FUEL PERFORMANCE
MODELS FEEDBACK PARAMETERS*

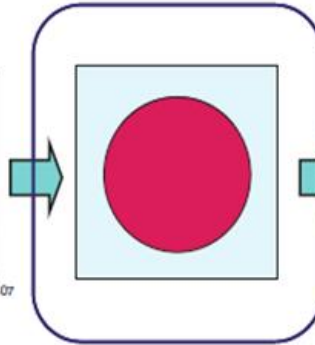
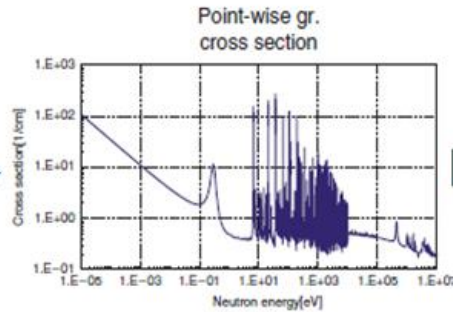
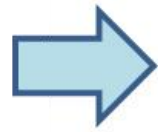
To recall:

The multi-physics simulation tools could be divided in three groups (categories): pre-traditional (simplified), traditional, and novel.

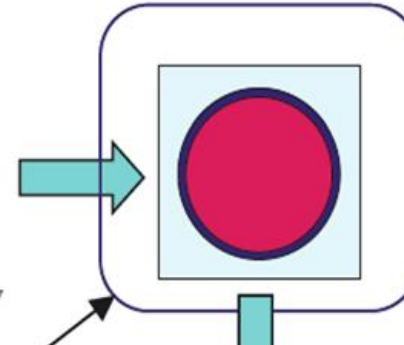
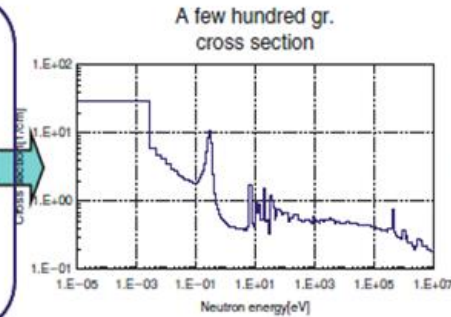
TRADITIONAL - SUMMARY

- » The discretization of the variables in the neutron transport equation is performed in a quite coarse and approximate way:
 - **two-energy groups** (or few groups) for treatment of energy variable;
 - **coarse-mesh** nodal or improved finite-difference or finite element method for treatment of spatial variable;
 - **diffusion approximation** for angular dependence;
 - using **nodal nuclear data** (library of parameterized nodal equivalent parameters).
- » The spatial neutronics/thermal-hydraulic coupling is performed on assembly/channel level.
- » Simplified fuel rod models are utilized as part of the system or sub-channel thermal-hydraulic codes.

TRADITIONAL - SUMMARY

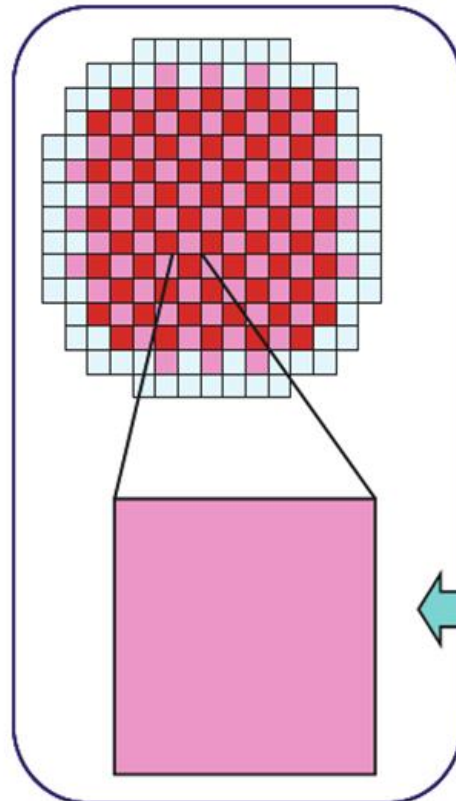
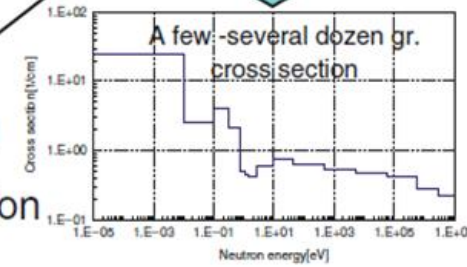


Resonance calculation

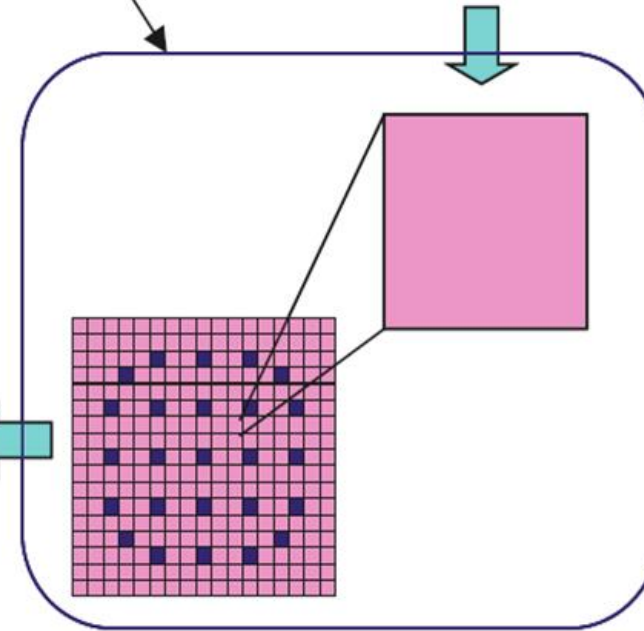
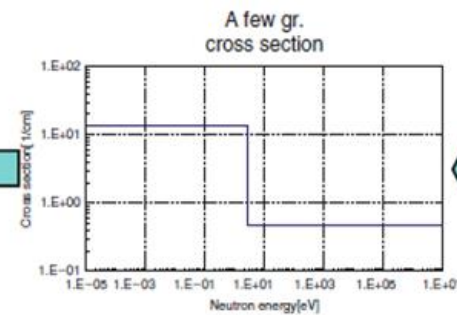


Pin-cell calculation

Assembly calculation

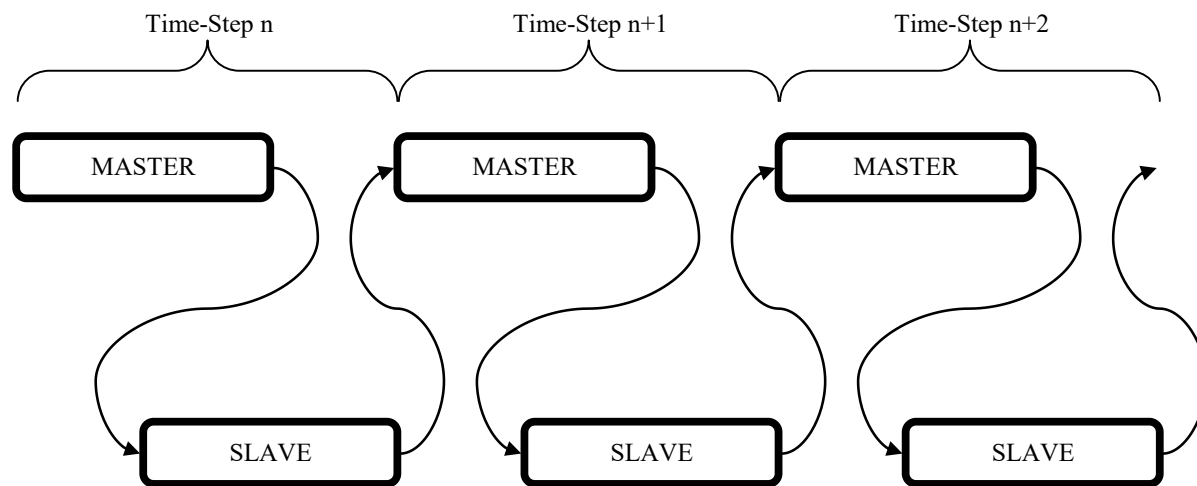


Core calculation

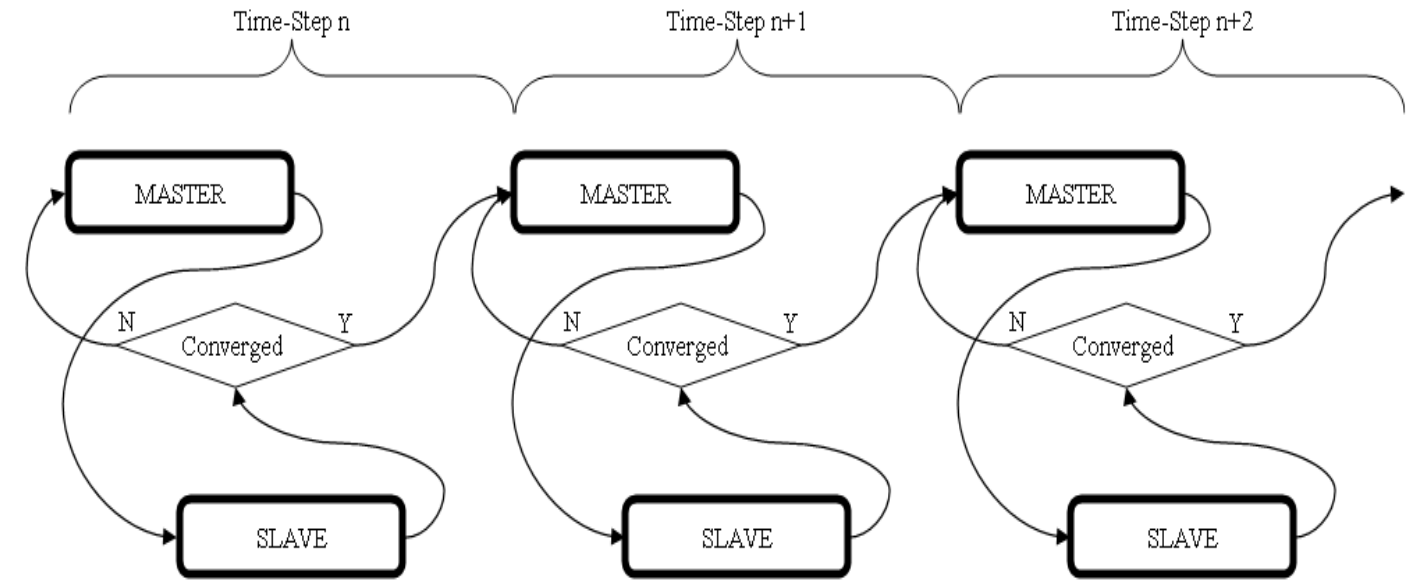


TRADITIONAL - SUMMARY

- » The evaluation of local safety related parameters on pin/sub-channel level for the hot assembly/channel is performed via pin-power reconstruction
- » The temporal neutronics/thermal-hydraulic coupling is performed in the operator-splitting approach using fixed point iterations for steady state simulations and sequential feedback parameter exchange (explicit coupling) at each time step for transient simulations.



Explicit temporal coupling



Fixed point iteration for temporal coupling

NOVEL MULTI-PHYSICS MODELING OF NUCLEAR REACTORS

- » The efficient and safe operation of current reactor fleet and the design of next generation reactor concepts along with the continuing computer technology progress have stimulated the development, qualification, and application of first principles high-fidelity multi-physics methodologies for reactor analysis.
- » The current efforts have been focused on extending the analysis capabilities by coupling models, which simulate different phenomena or different reactor components, as well as on refining the scale and level of detail of coupling.
- » In terms of improving the model capability for design and safety assessment the prediction of the fuel rod response is being based on first principles high-fidelity modeling of local parameters.
- » Subsequently, the current developments are towards performing whole core high-fidelity multi-physics calculations on pin/sub-channel level and even on sub-pin level in order to understand and predict key aspects of fuel and cladding performance.
- » Such capabilities allow determining radial, axial and azimuthal flux distribution, energy deposition, coolant and fuel temperature distributions for different pin cells/sub-channels at different core locations.

NOVEL MULTI-PHYSICS MODELING

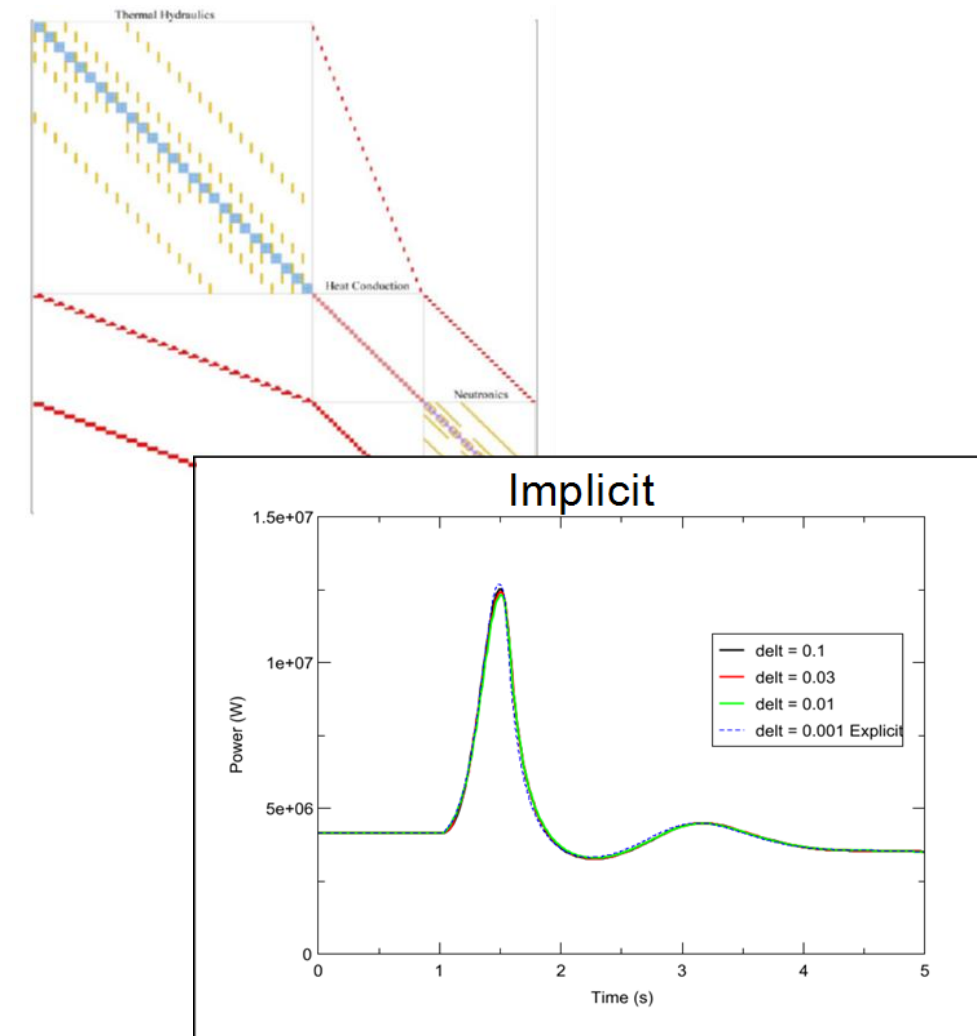
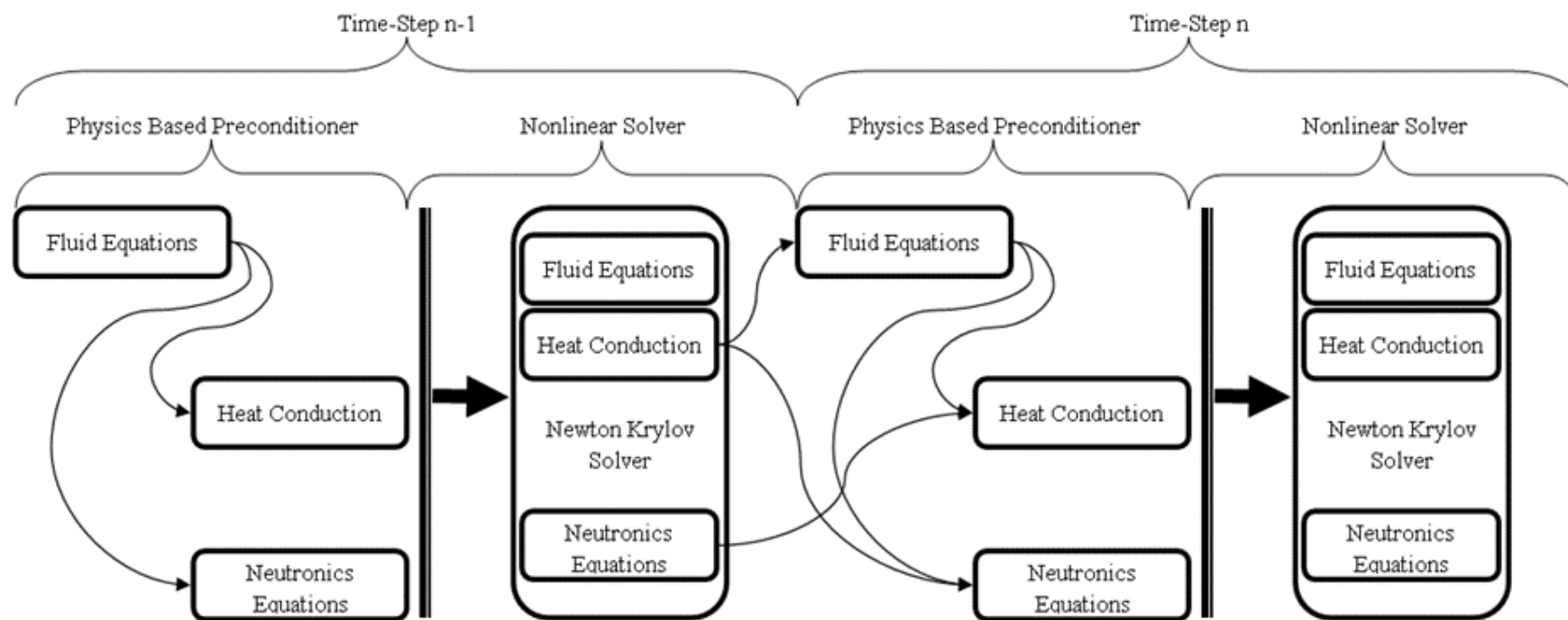
Common features of more physics-based high-fidelity simulations:

1. For the deterministic neutronics methods **two-energy group diffusion** neutronics solvers are replaced with multi-group transport solutions.
2. Continuous energy Monte Carlo neutronics codes are coupled with thermal-hydraulic solvers for reference coupled multi-physics solutions.
3. The thermal-hydraulic solvers are based on sub-channel, porous media or CFD solutions.
4. The coupling is done spatially on pin/sub-channel level and even on within pin heterogeneity level, which requires the implementation of improved and flexible coupling methodologies.
5. Such refined coupled simulations can be performed in direct and embedded manner utilizing multi-level coupling schemes.

NOVEL MULTI-PHYSICS MODELING

More physics-based high-fidelity simulations:

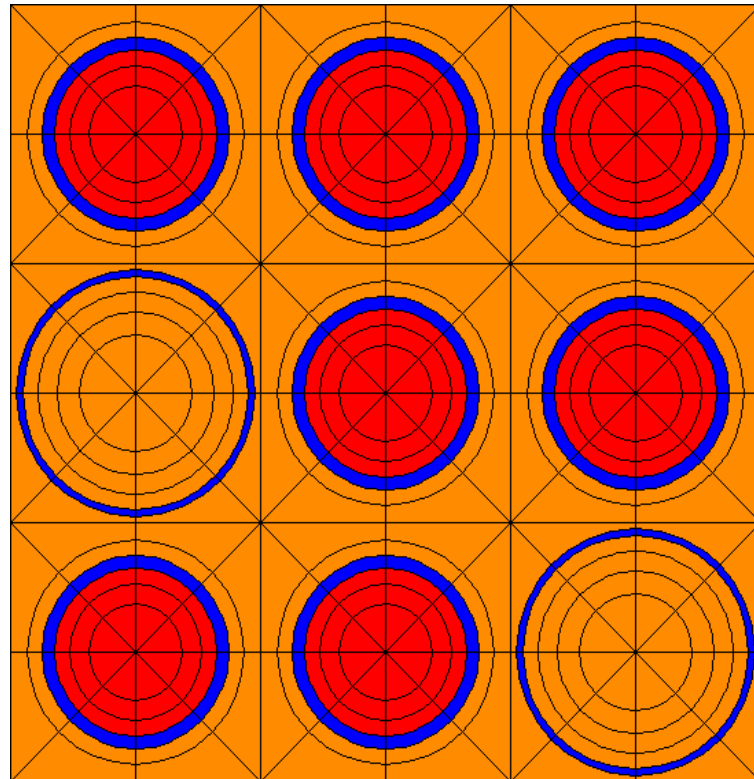
1. High Performance Computing is efficiently utilized via modern and user-friendly simulation platforms and parallel computing.
2. Fully implicit time-integration methods for simultaneous solution of the coupled non-linear system are implemented.



NOVEL MULTI-PHYSICS MODELING

More physics-based high-fidelity simulations (cont.):

3. High-fidelity neutronics, thermal-hydraulics, and fuel rod models (based on fuel performance codes) are coupled in reactor core simulations.
4. Full core high-fidelity multi-physics calculations on pin/sub-channel level and even on sub-pin level are needed to understand and predict key aspects of fuel and cladding performance.



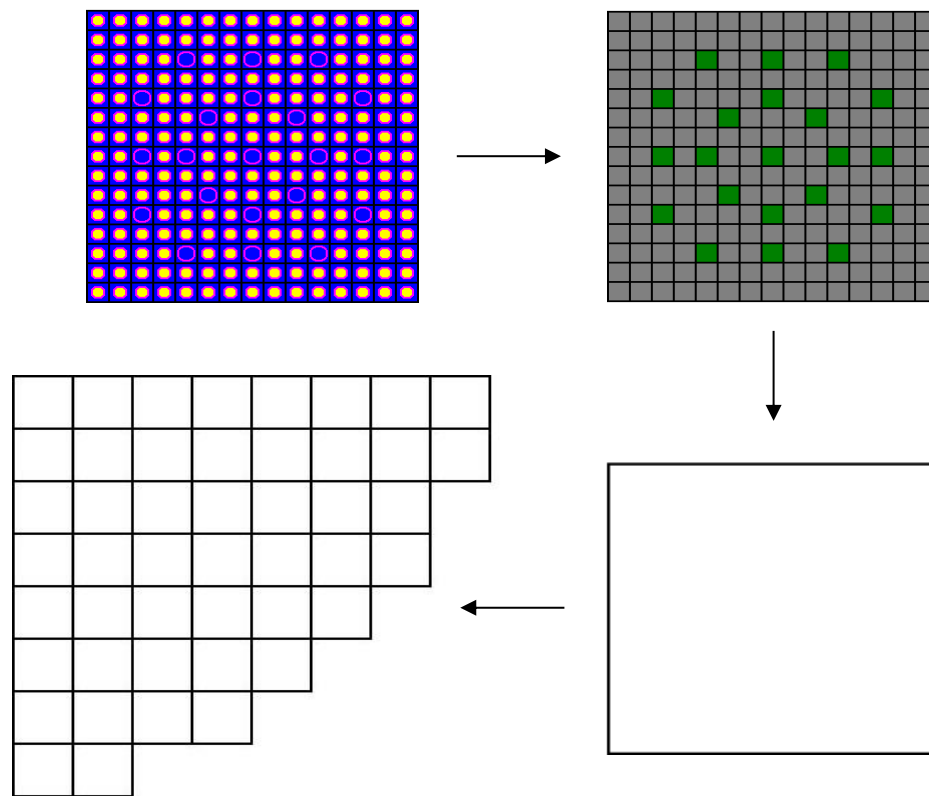
Level of resolution required for modeling hydride distribution in the cladding of fuel pins

HIGH-FIDELITY NEUTRONICS

HIGH-FIDELITY NEUTRONICS

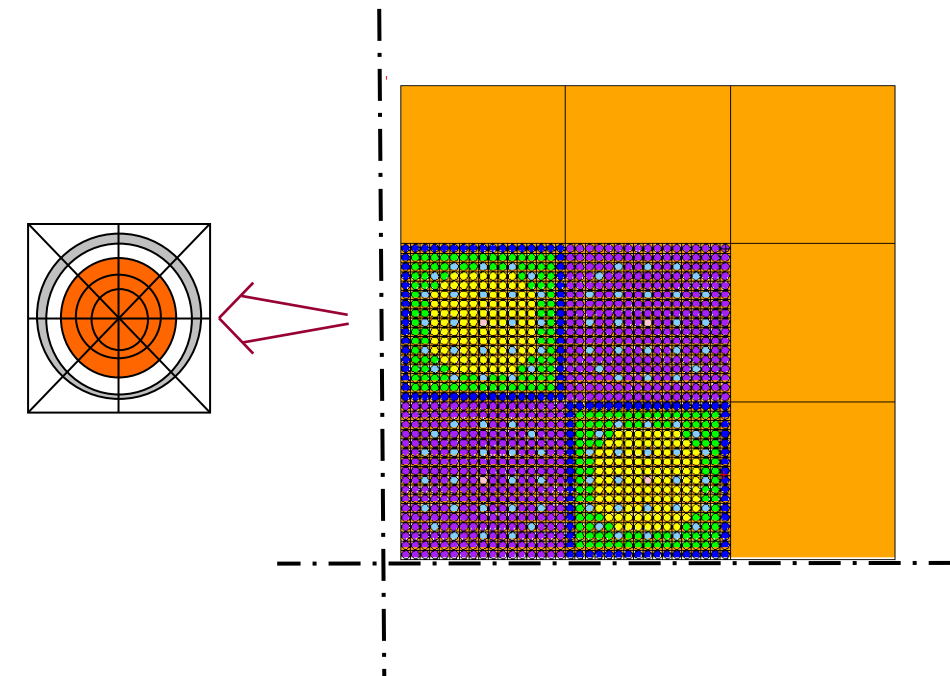
Conventional Approach (PARCS):

1. Multi-step calculations involving homogenization and group condensation.
2. Neutron diffusion approximation for core calculation.

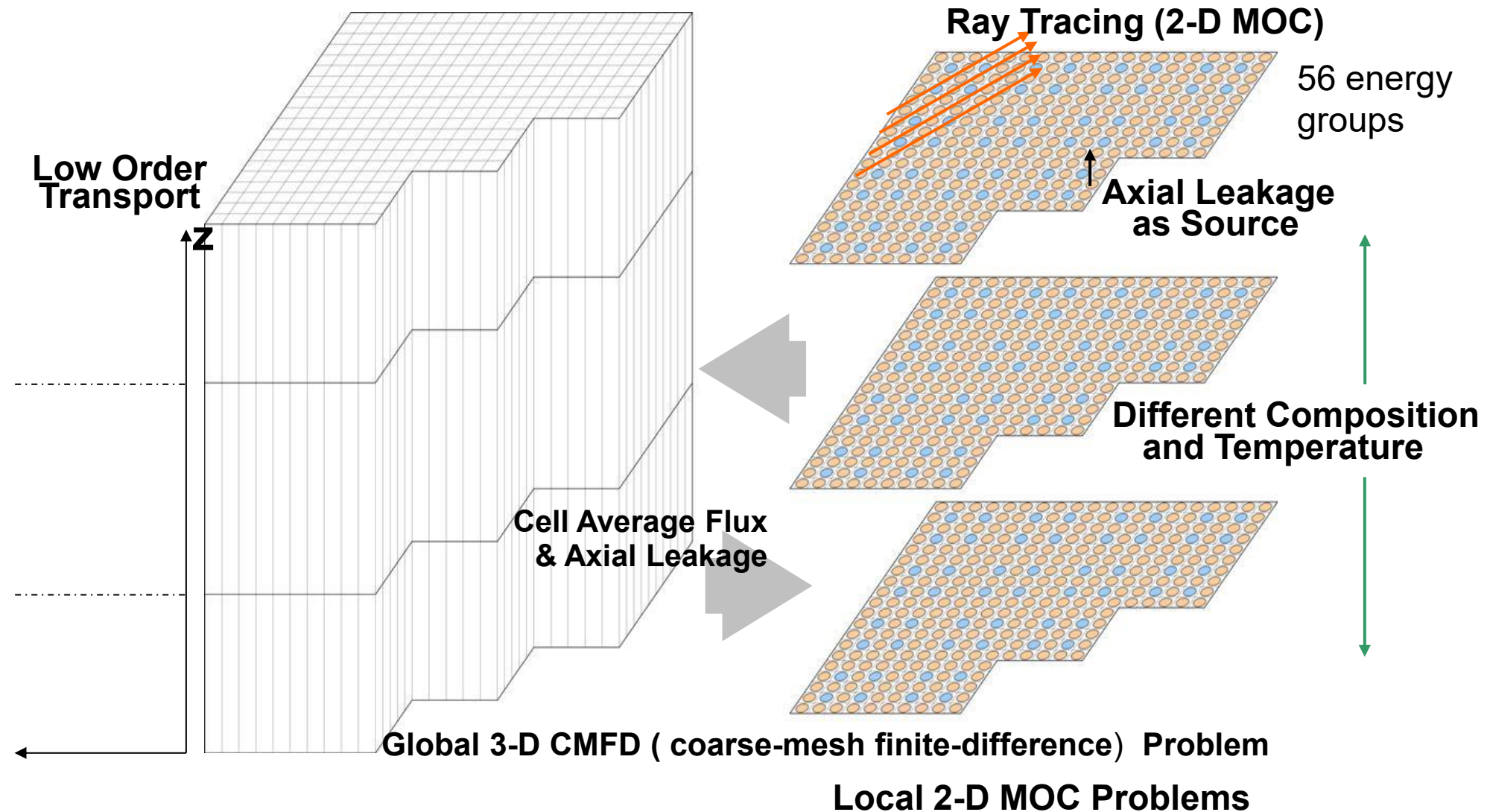


Next Generation Method (MPACT):

1. Direct full-core integral transport calculation with explicit representation of local heterogeneity / in-line resonance treatment (sub-group method).
2. No homogenization.
3. Sub-pin level T/H Feedback through Coupling with Subchannel or CFD.

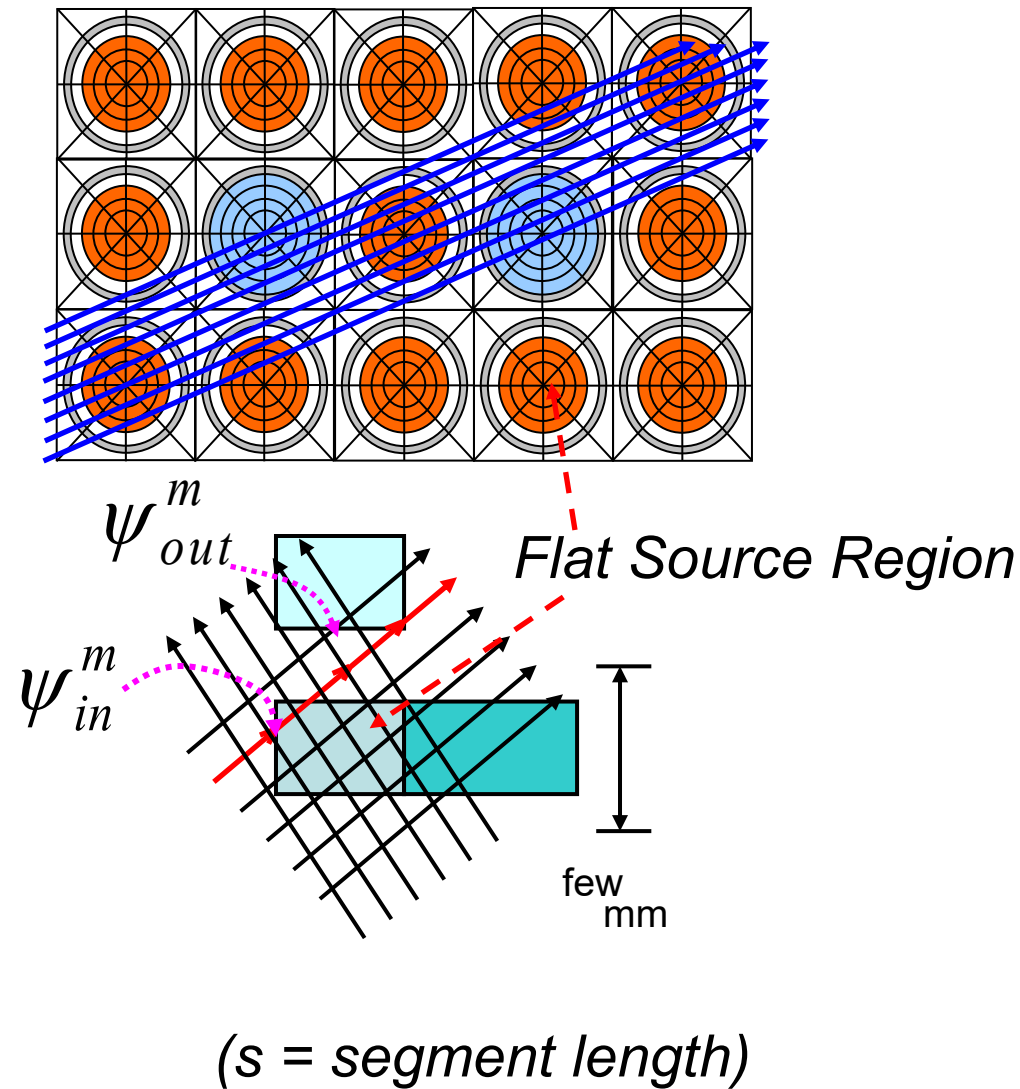


2D/1D FORMULATION FOR DIRECT FULL CORE TRANSPORT



$$\left(\eta_m \frac{\partial \phi_g^{m,n}}{\partial x} + \varepsilon_m \frac{\partial \phi_g^{m,n}}{\partial y} \right) + \Sigma_{tg} \phi_g^{m,n} = \frac{1}{4\pi} \left(\chi_g \psi^n + \sum_{g'=1}^G \Sigma_{gg'} \phi_{g'}^n \right) - \frac{\mu_m}{h_z} (\phi_{gT}^m - \phi_{gB}^m) + \frac{1}{4\pi} S_{tr}^{g,n}$$

RAY TRACING AND METHOD OF CHARACTERISTICS



Neutron balance along ray

$$\frac{d\psi(\xi)}{d\xi} + \Sigma\psi(\xi) = q ; \quad \psi(0) = \psi_{in}$$

Assume constant cross-sections
and flat source
within a micro region

$$\psi_{out}^m = \psi_{in}^m e^{-\Sigma s} + \frac{q}{\Sigma} (1 - e^{-\Sigma s})$$

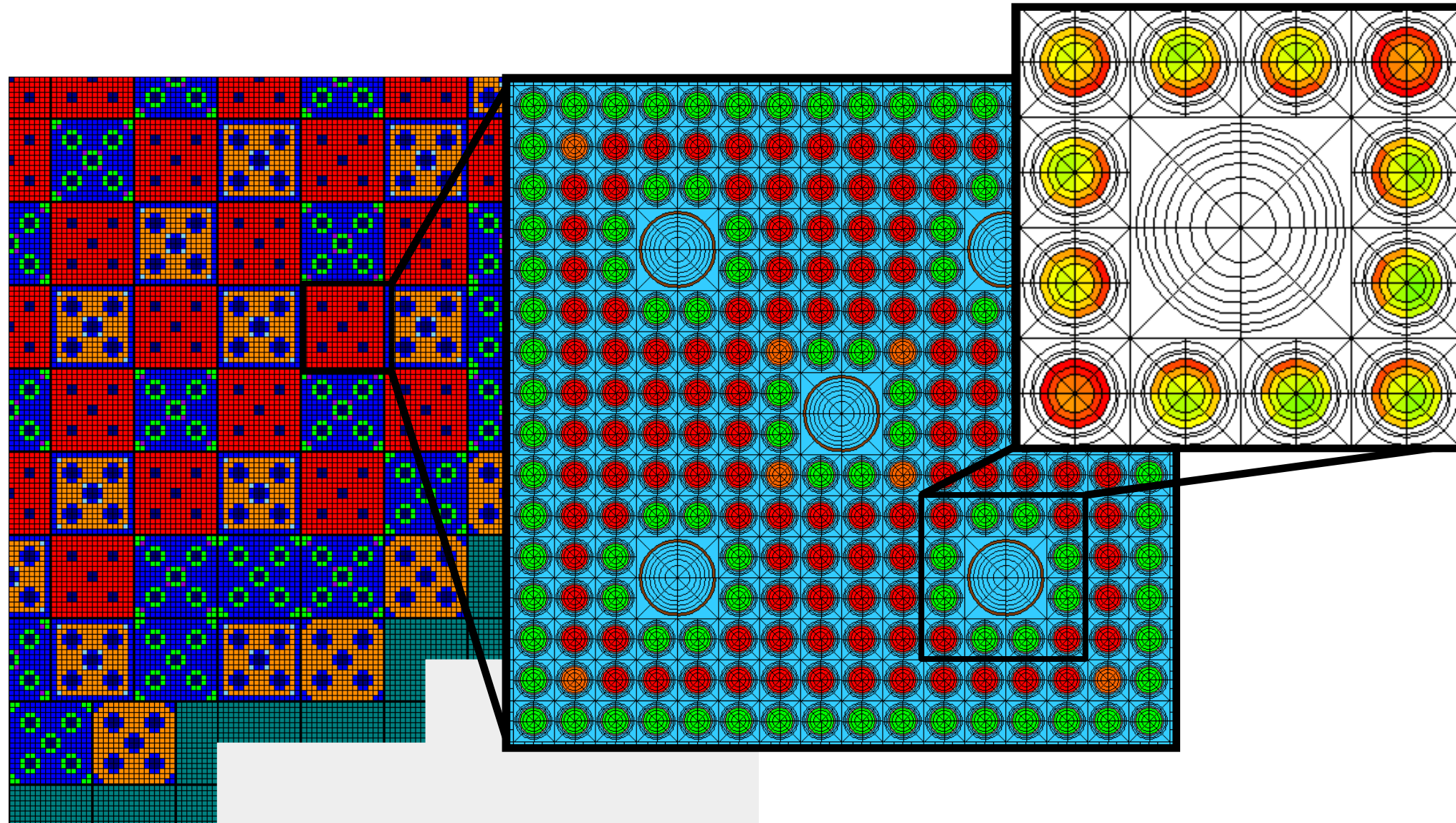
$$\bar{\psi}_m = \frac{\psi_{in}^m - \psi_{out}^m}{\Sigma s} + \frac{q}{\Sigma}$$

Segment
average
angular flux

$$\phi = \sum_m \omega_m \bar{\psi}_m$$

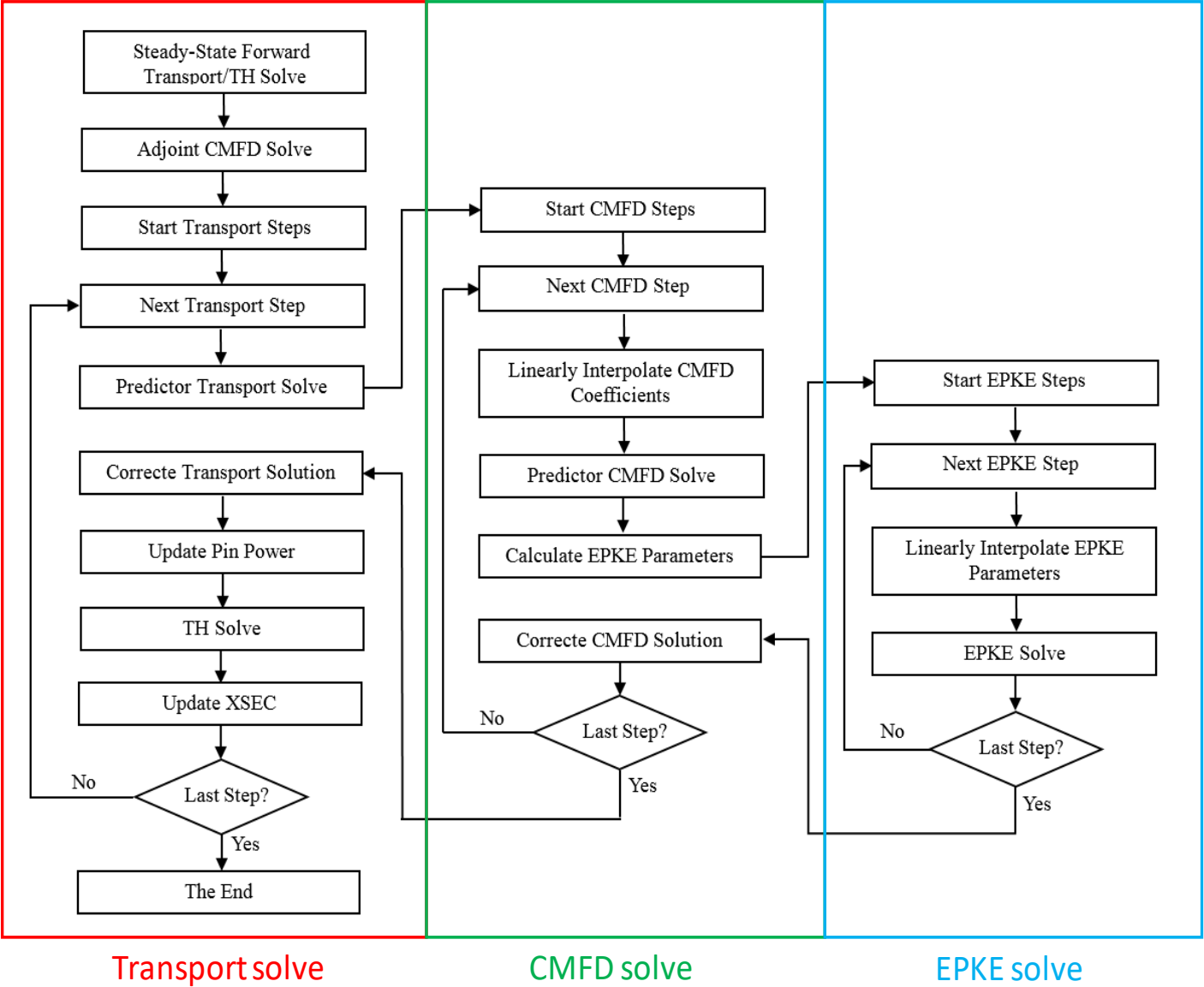
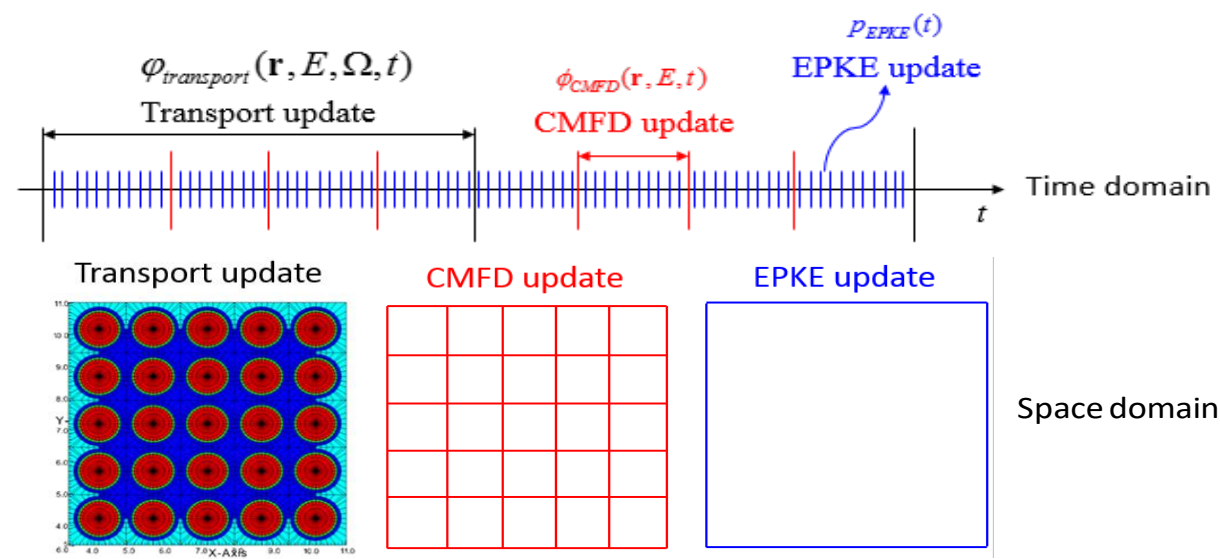
Scalar flux

FULL CORE “PIN RESOLVED” TRANSPORT

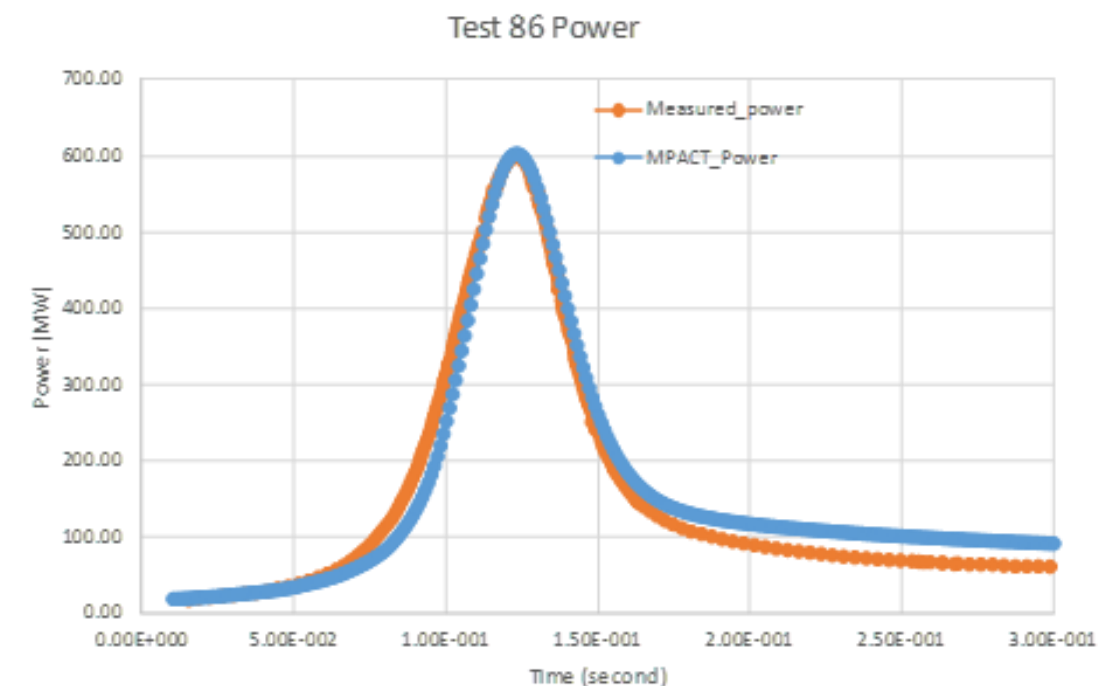
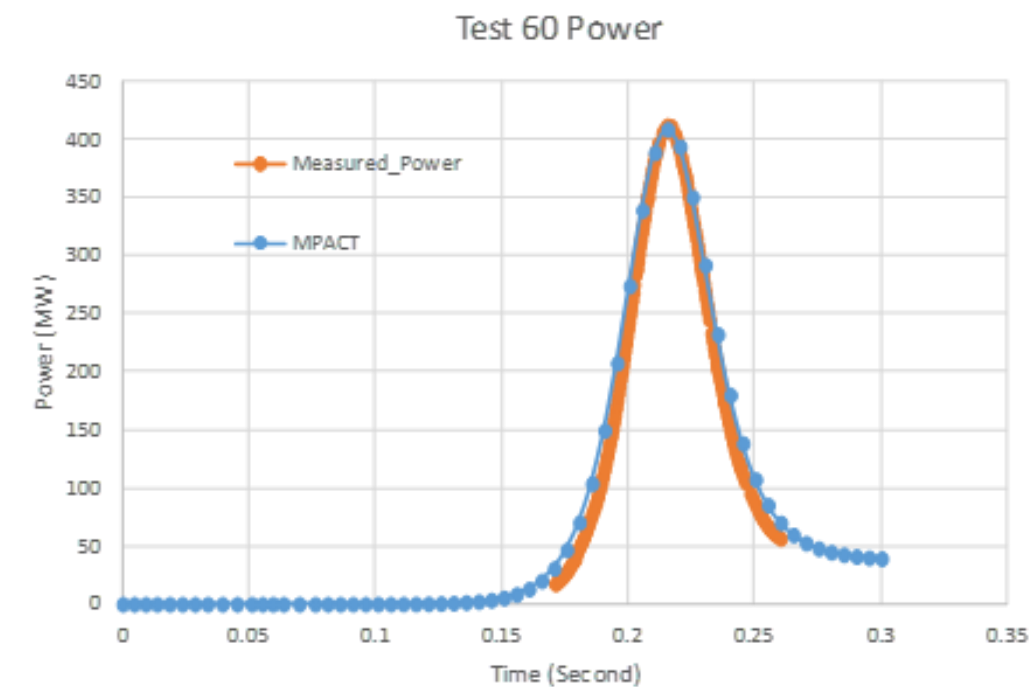
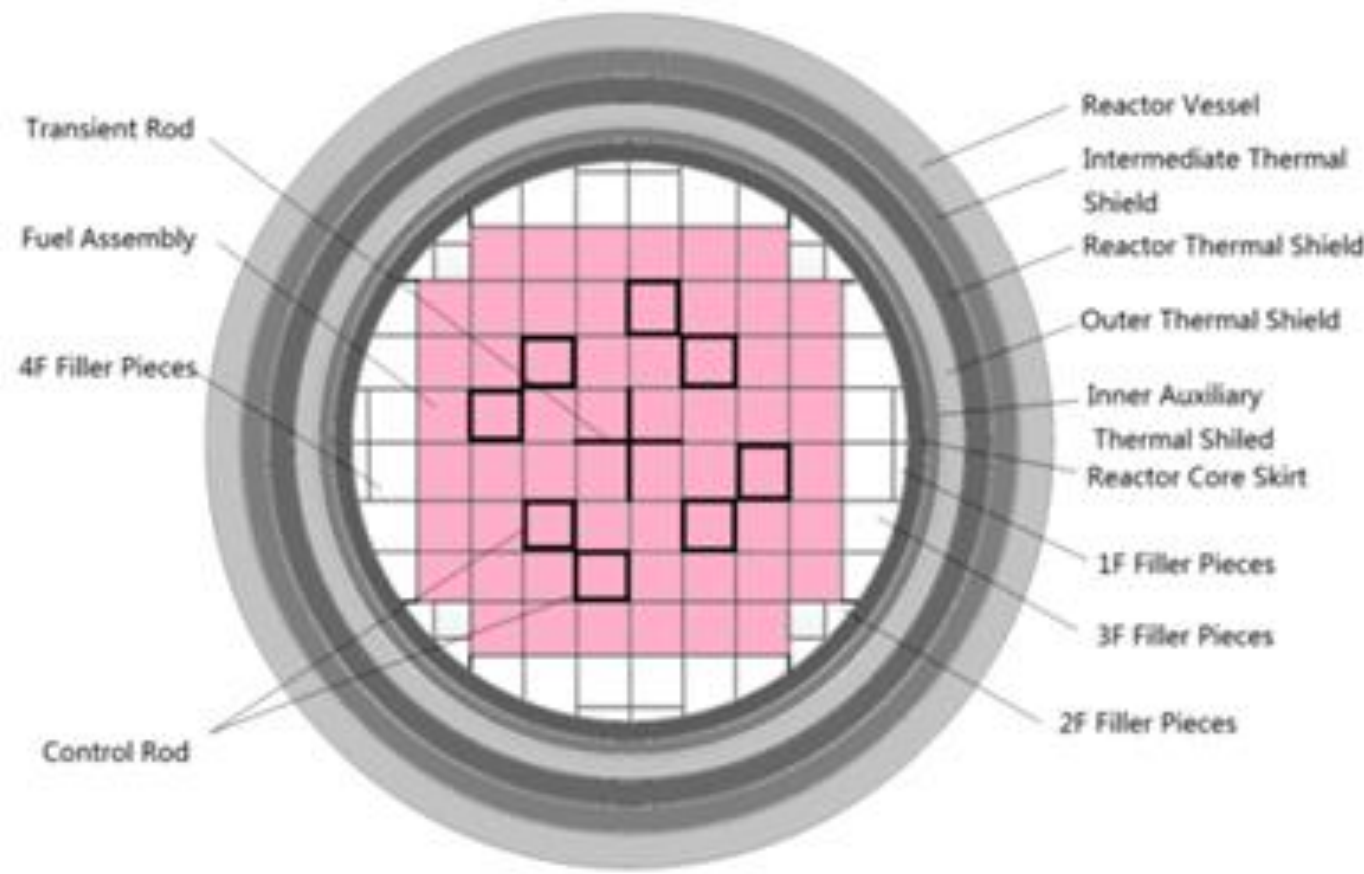


TRANSIENT-MULTI-LEVEL METHOD FOR MPACT

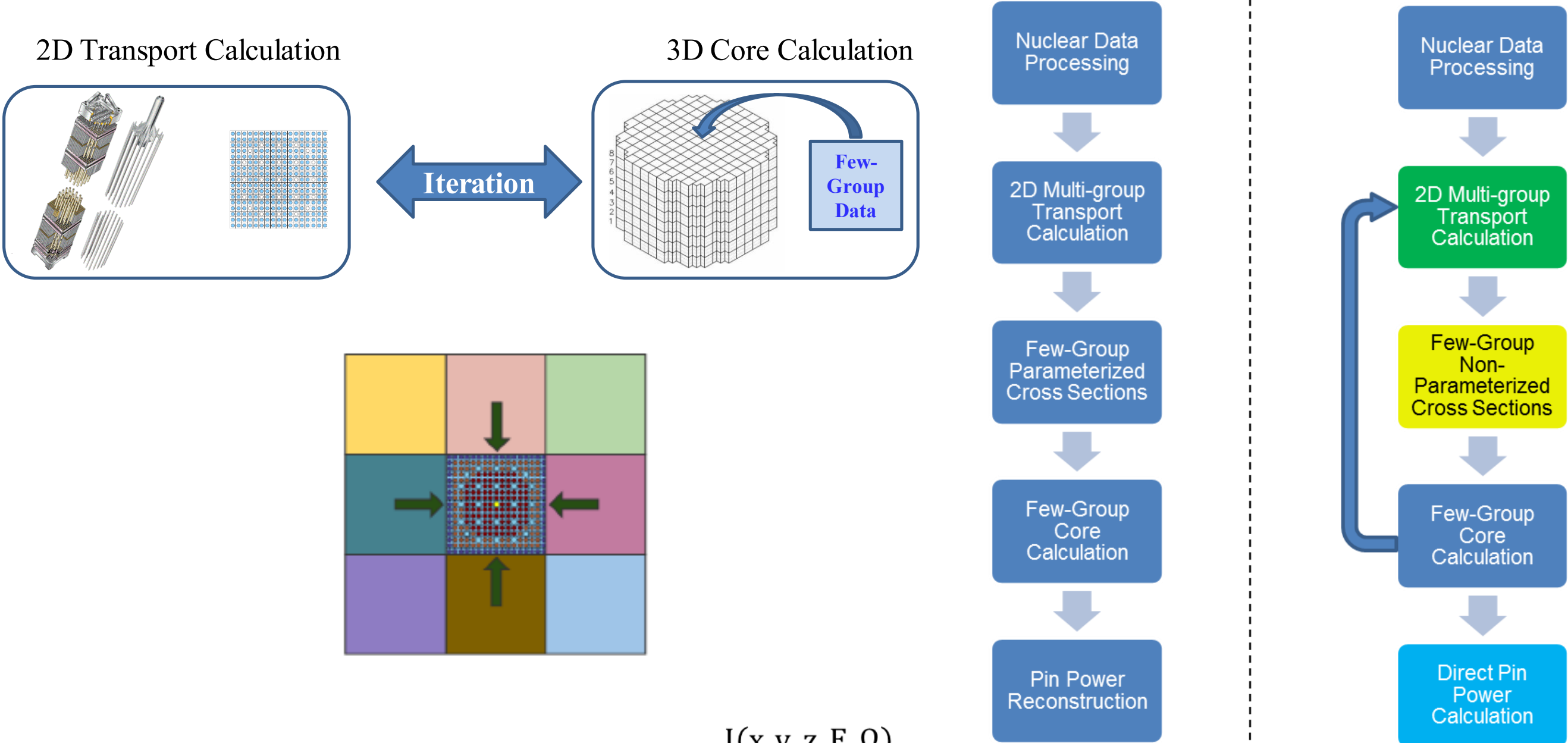
Combines elements of the Predictor-Corrector Quasi-static method (PCQM) and Multi-grid amplitude function (MAF) method.



TRANSIENT-MULTI-LEVEL METHOD FOR MPACT



ITERATIVE TRANSPORT-DIFFUSION METHOD

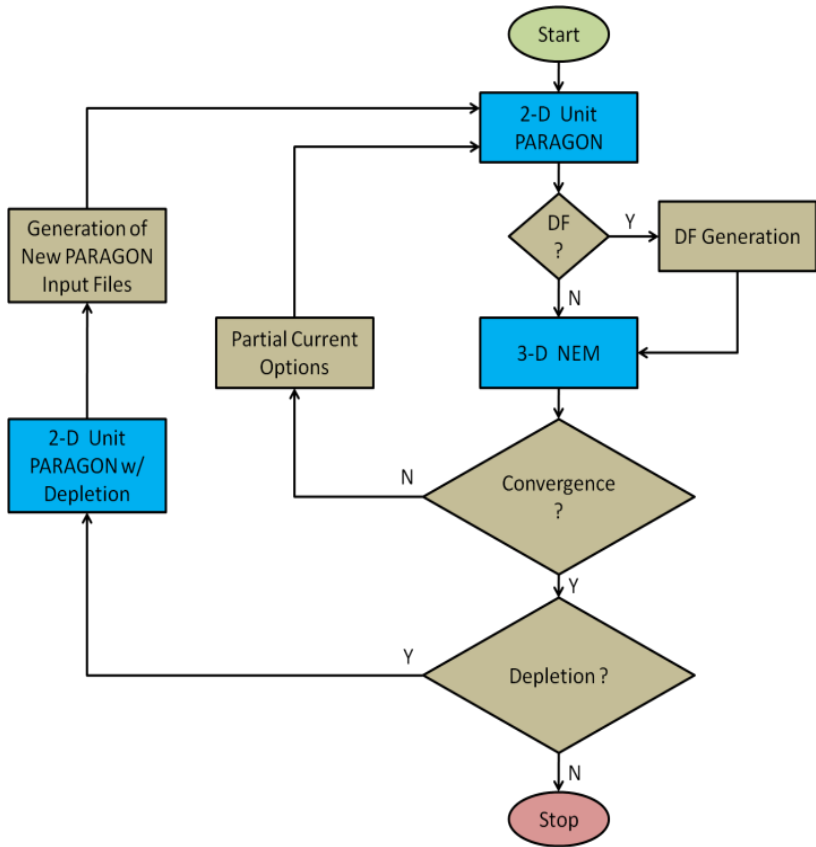


Node Partial Current Shape $(x, y, z, E, \Omega) = \frac{J(x, y, z, E, \Omega)}{\int dx \int dE \int d\Omega J(x, y, z, E, \Omega)}$

ITERATIVE TRANSPORT-DIFFUSION METHOD

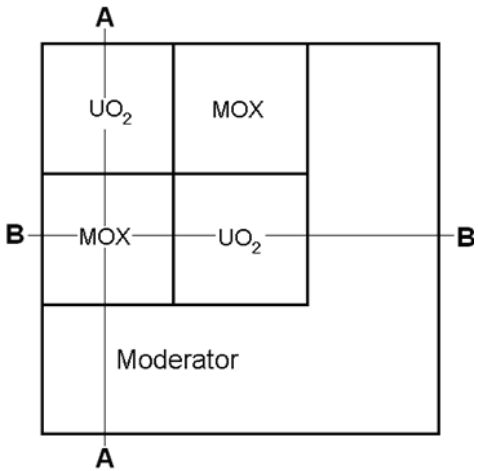
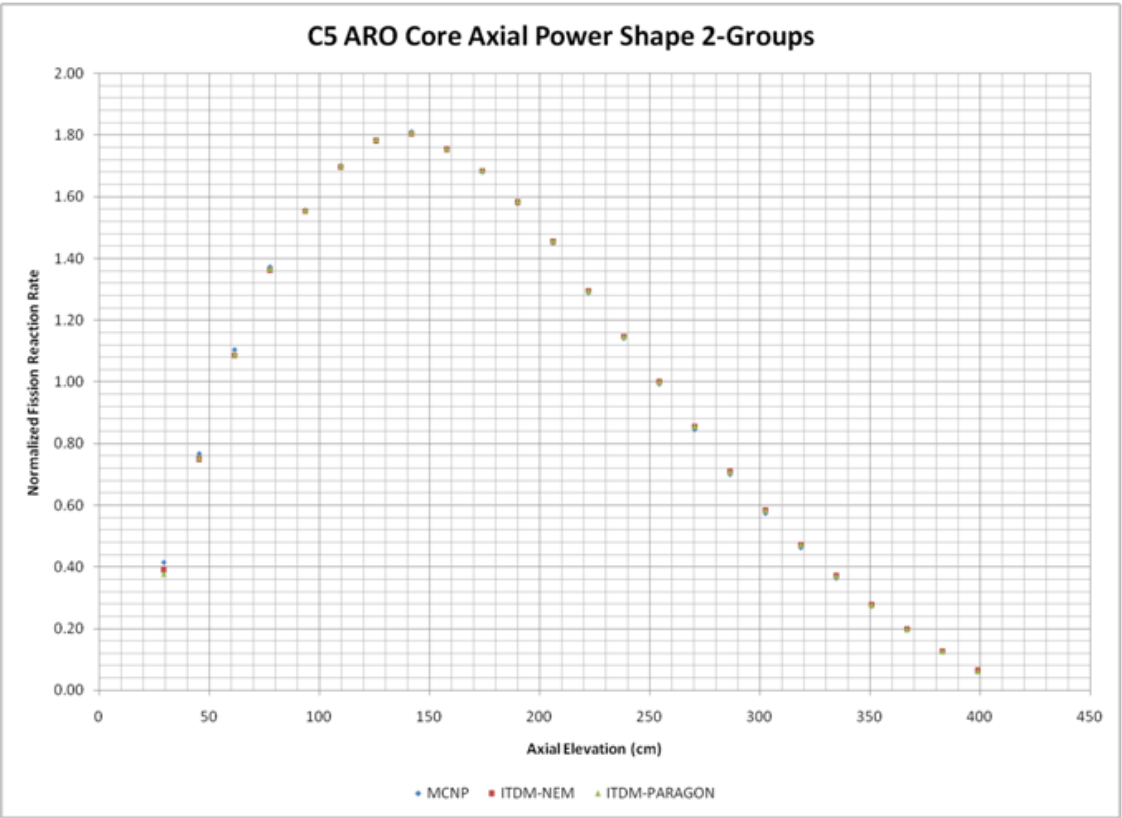
Embedded (non-linear iteration) 3-D neutronics methods:

- Method of Characteristics (MOC)
- Hybrid MOC/MCNP
- Collision Probabilities
- S_N , SP_3 , P_N , Diffusion
- Issues:
 - Axial leakage modeling
 - Stability and convergence
- Naturally lending themselves to parallel computing



Flow-chart of Iterative Transport Diffusion Methodology

ITDM predictions of axial power distribution for PWR MOX mini-core



Radial layout of PWR MOX mini-core

HIGH-FIDELITY THERMAL-HYDRAULICS

LOW FIDELITY VS. HIGH FIDELITY TH

- System & Subchannel TH (STH) codes $\neq$ CFD codes
- Originating principles and fluid balance equations are the same for STH and CFD codes, however....
- STH codes do not calculate 'profiles' (e.g. velocity) in cross sections
- STH code equations are solved within solid boundaries; CFD code equations are solved within the fluid domain
- STH codes need a 'huge' number of constitutive-empirical correlations
- CFD codes calculate the turbulence
- CFD codes calculate fluid performance at the scale of 10^{-4} m
- STH codes are 'RA-approved' computational tools in nuclear reactor safety
- Coupling needed between STH and CFD codes to solve complex multi-scale problems in nuclear reactor safety

SUBCHANNEL APPROACH VS. RANS

<i>RANS (CFD Codes)</i>	<i>System Codes</i>	<i>Subchannel Codes</i>
Gravitation	(+)	(+)
Transverse flow between subchannels due to lateral pressure gradients (diversion crossflow)	(-)	(+)
Pressure Losses: frictional losses head losses interfacial drag forces	(+) (+) (+)	(+) (+) (+)
Lateral exchange between channels / subchannels due to molecular and turbulent diffusion in axial flow (turbulent mixing)	(-)	(+)
Void drift	(-)	(+)
Lateral exchange between subchannels due to molecular and turbulent diffusion in swirl flow in a horizontal plane (turbulent mixing in the transverse momentum equation)	(-)	(-)
Lateral exchange between subchannels due to centrifugal forces in swirl flow in a horizontal plane	(-)	(-)
Transverse flow between subchannels due to other flow patterns created by spacers/spacer vanes (directed flow)	(-)	(- / +)
Lift force (Magnus effect on bubbles and droplets; relative velocity and rotation in velocity field of continuous phase)	(-)	(-)
Turbulent dispersion force (Diffusive bubble movement due to turbulence in the continuous phase)	(-)	(-)
Virtual mass force	(-)	(-)
Wall lubrication force	(-)	(-)
Entrainment and deposition of droplets in annular flow	(- /+)	(+)

FEATURES OF SYSTEM / SUBCHANNEL TH CODES

System TH Code Capabilities

- Empirical Constitutive Eqs & Partial Derivative 6-Eqs
- Porous Media Approach (Not Open Media, Not CFD, Not DNS)
- 1D, pseudo 3D TH modeling is possible
- ‘Fully’ Assessed Code: Accuracy Quantification & Uncertainty Evaluation
- ‘System’ TH Code

Capabilities to be assessed

- Interfacial Area Transport
- 3-D Porous Media
- CFD-coupling
- Individual Fuel Rod Sub-channel Analysis

Subchannel TH Code Capabilities

Empirical Constitutive Eqs & Partial Derivative **9-Eqs**
Subchannel Media Approach (Not Open Media, Not CFD, Not DNS)

2D, pseudo 3D TH modeling is possible

‘Fully’ Assessed Code: Accuracy Quantification & Uncertainty Evaluation

Not System TH (**BCs needed**)

Fuel rod modeling / Local safety limits

Interfacial Area Transport

CFD-coupling (**high-to-low model information**)

Full Two-phase CFD / DNS (Direct Numerical Simulation) May Need Decades

FEATURES OF SYSTEM / SUBCHANNEL TH CODES

Main Goals for System / Subchannel Codes

- Nuclear Power Plant (NPP) Design
- NPP Operation
- NPP Modification
- NPP Safety Evaluation
- NPP Licensing

Range of application is much broader, including

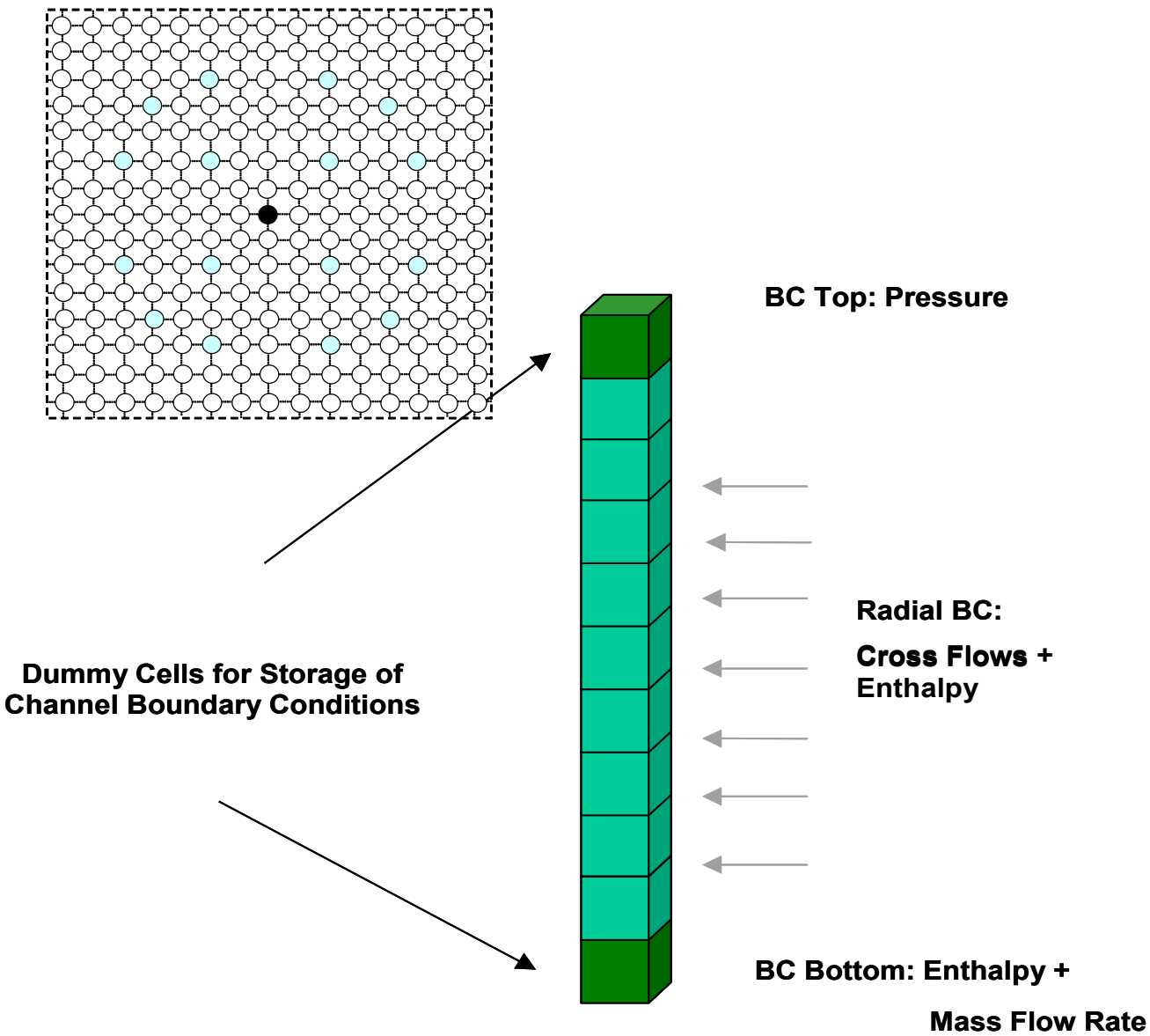
- Analysis of Experimental Data (also part of the Code Qualification Process)
- Interpretation or Understanding of Complex Phenomena
- Operator Training
- Qualification of Simulators and Plant Analyzers
- Analysis of Operational Transients (also part of the Code Qualification Process)
- Non-Nuclear Application

TRADITIONAL CORE TH M&S (EXAMPLE: SUBCHANNEL ANALYSIS CODES)

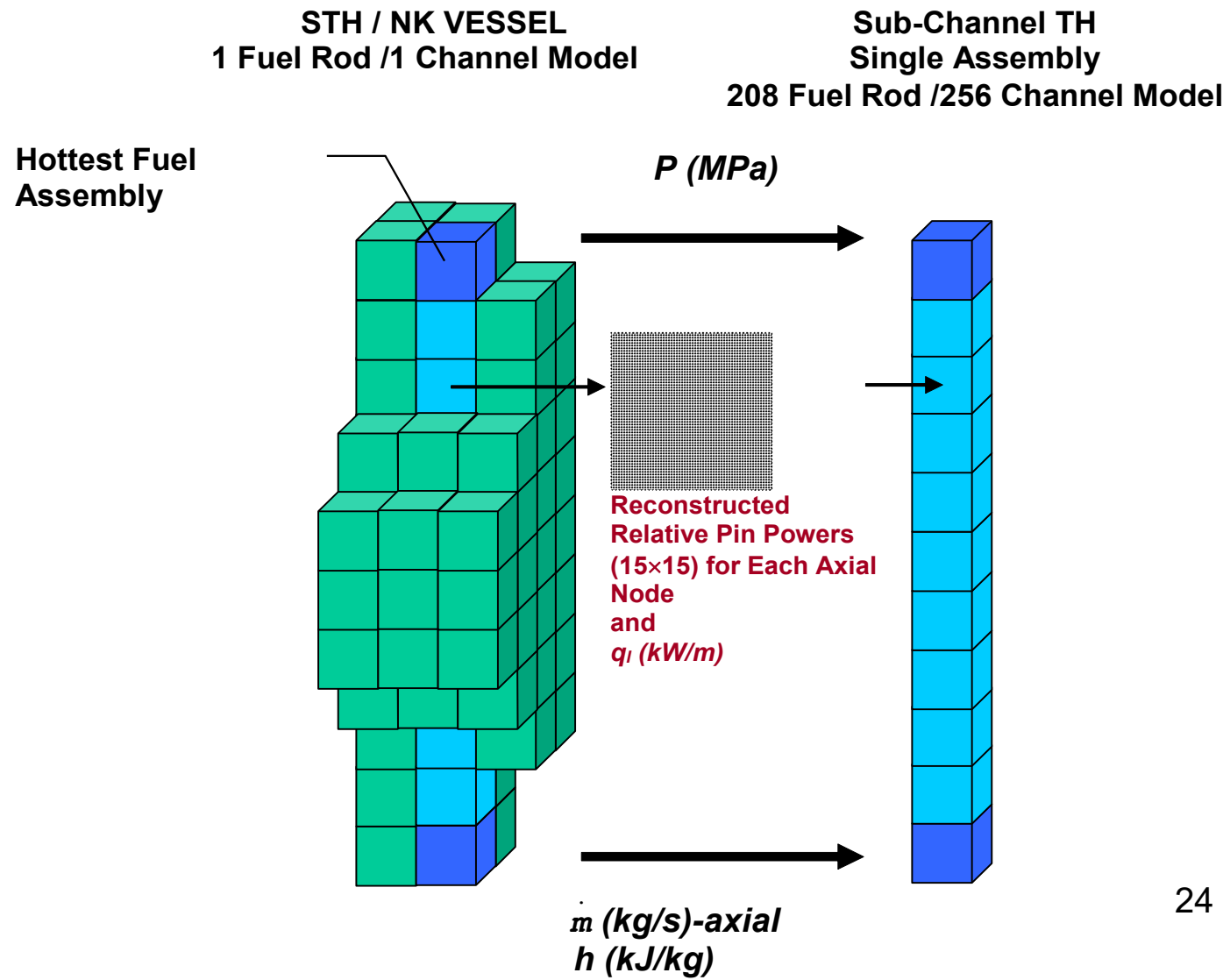
- Traditionally, thermal-hydraulic subchannel codes have been used:
 - either for standalone rod bundle M&S on a subchannel level,
 - or for coupled TH with neutronics reactor core M&S on a fuel assembly level.
- As part of integrated safety capability developments, TH subchannel models have been utilized for both hot fuel assembly and whole core analysis in one-way coupling using pin-power reconstruction techniques.

TRADITIONAL CORE TH M&S

Single Fuel Bundle



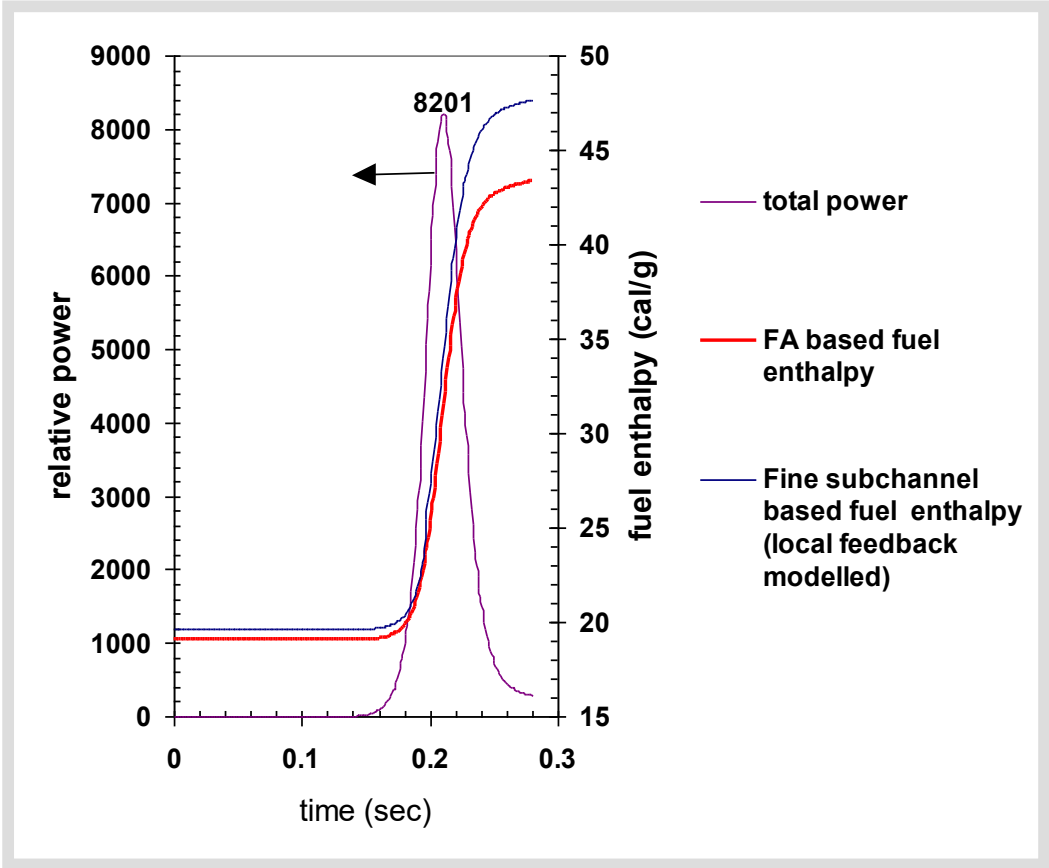
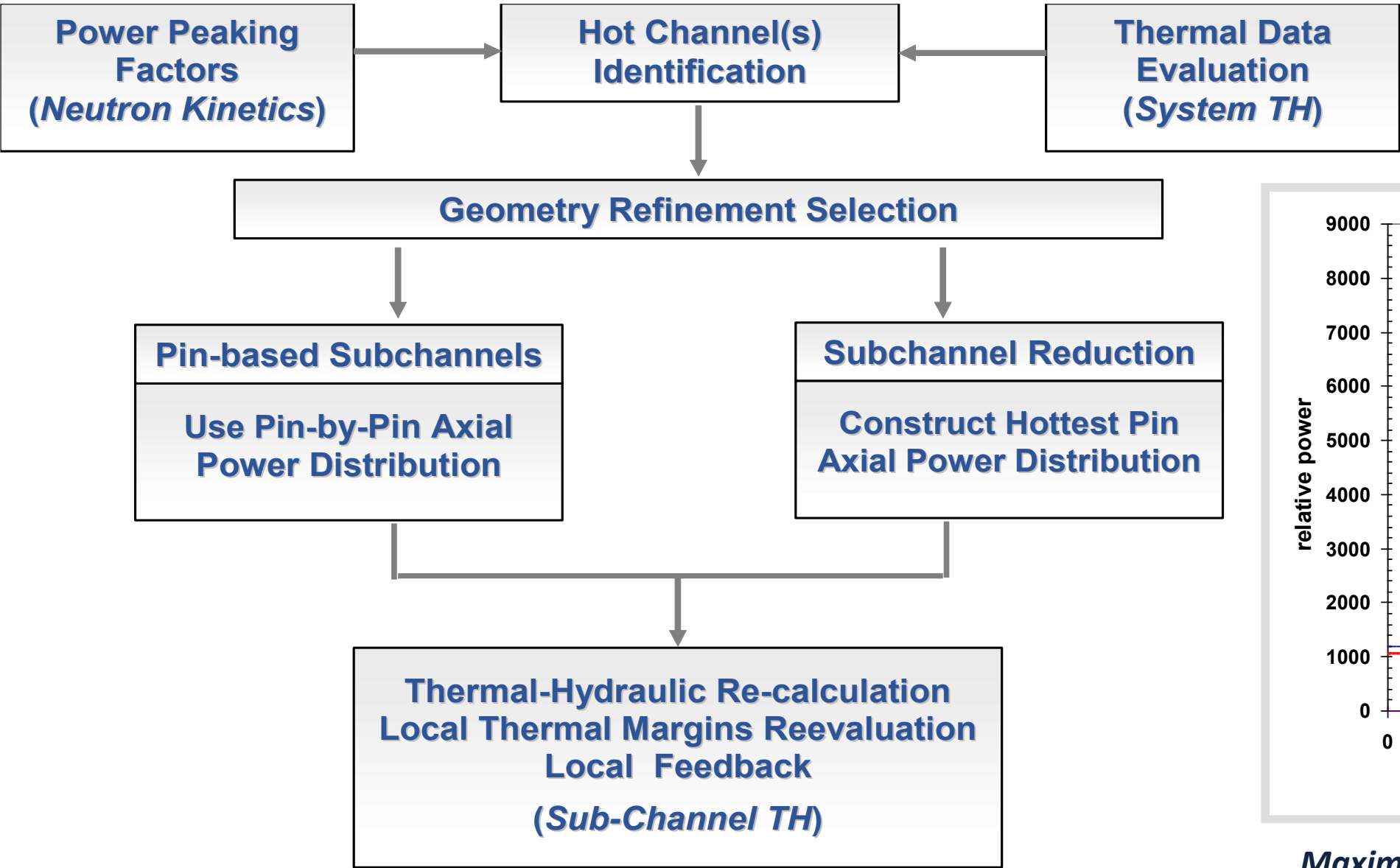
One-way coupling using pin-power reconstruction



Source: D. Zialetsev, et al. 2002

TRADITIONAL CORE TH M&S

Integrated Local Safety Evaluation



Maximum Total Power and Fuel Enthalpy REA 25

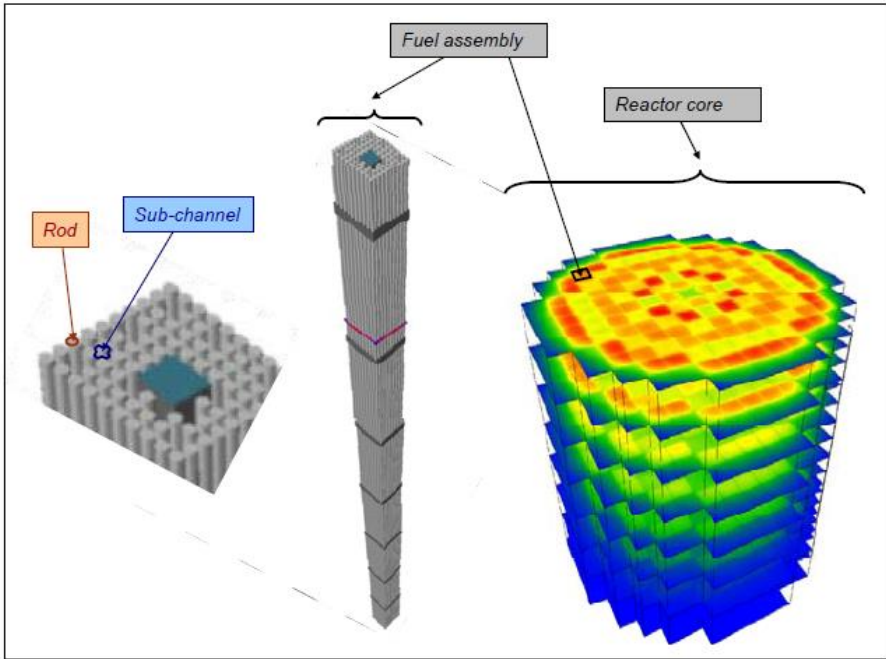
Source: D. Zialetsev, et al. 2002

TOWARDS NOVEL CORE TH M&S

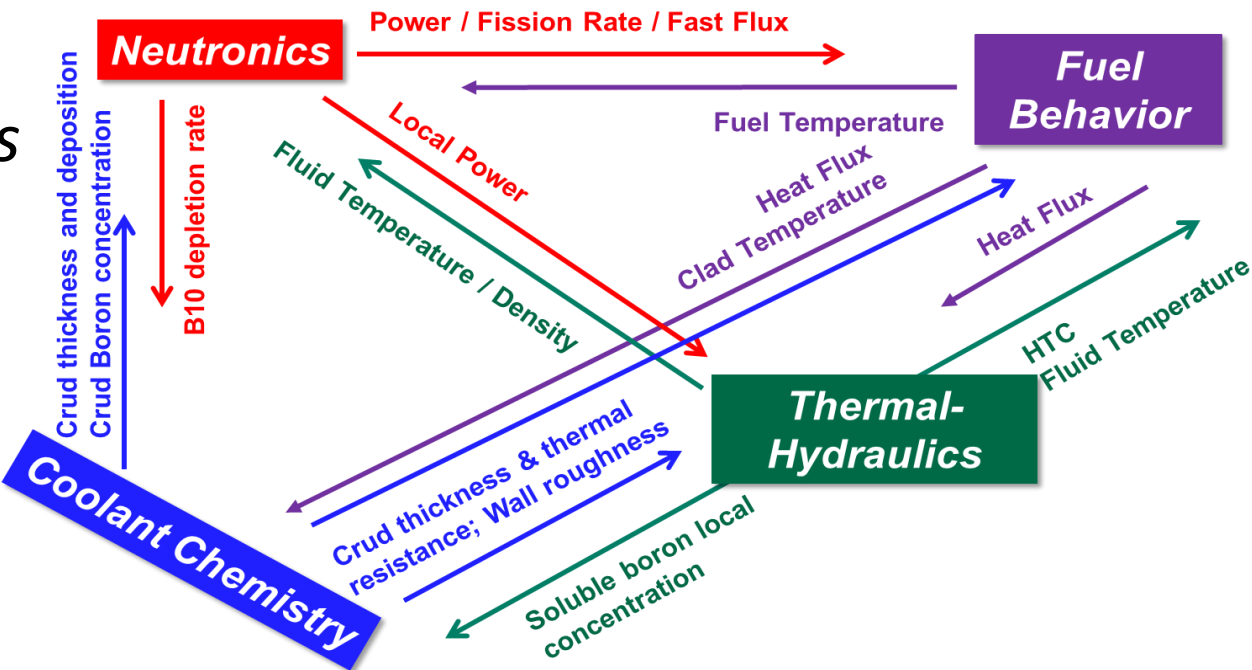
- TH subchannel codes are being coupled with neutron transport codes and fuel performance codes for high-fidelity reactor core M&S on a pin (sub-pin)/sub-channel level.
- !! To make these high-fidelity multi-physics codes accurate and efficient, developments are needed to improve the individual physics components as well as the coupling between them.

TOWARDS NOVEL CORE TH FOR MP M&S

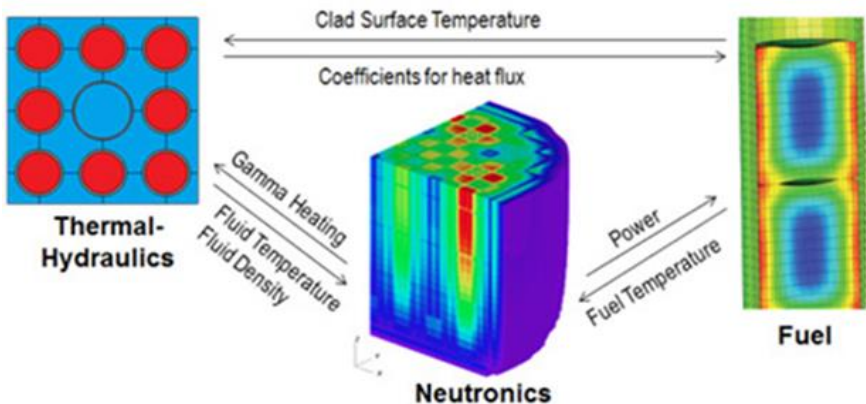
Pin/sub-channel level



Multi-physics coupling mechanisms in a LWR core (if no changes in geometry)



Multi-physics coupling mechanisms for short time transients in LWR



Source: RDFMG & CASL

TOWARDS NOVEL CORE TH M&S

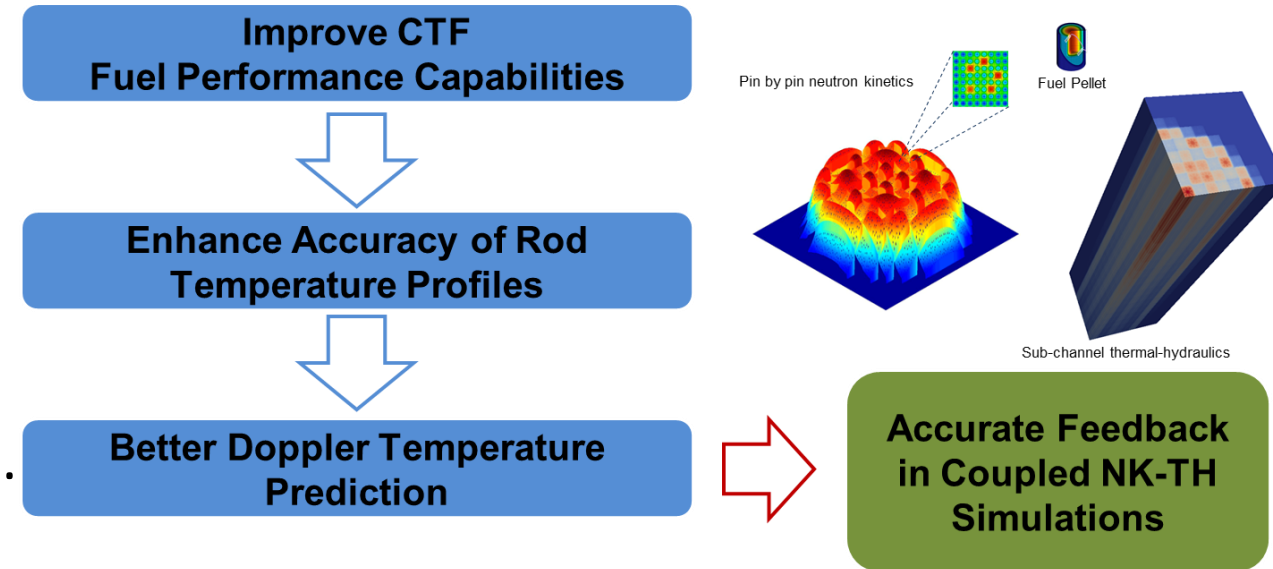
Improved Physics: CFD-informed models

- A physics-based approach which utilizes high-fidelity data to inform and calibrate a lower-fidelity simulation
- The goal is to increase the accuracy of predictions of quantities of interests using high-fidelity CFD models while preserving the efficiency of low-fidelity sub-channel core calculations
 - The development of such methodology should be supplemented by verification and validation activities
- The need of CFD-informed models in low-fidelity, low-resolution thermal-hydraulic codes:
 - CFD simulation resolution allows for capturing effects beyond the scope of a subchannel code calculation
 - Subchannel codes average flow characteristics within the mesh cell
 - Sub-channel codes maintain computationally-efficient simulation capabilities
 - Full-scale CFD simulations are not yet feasible, but selected models and correlations can be scaled for use in subchannel codes

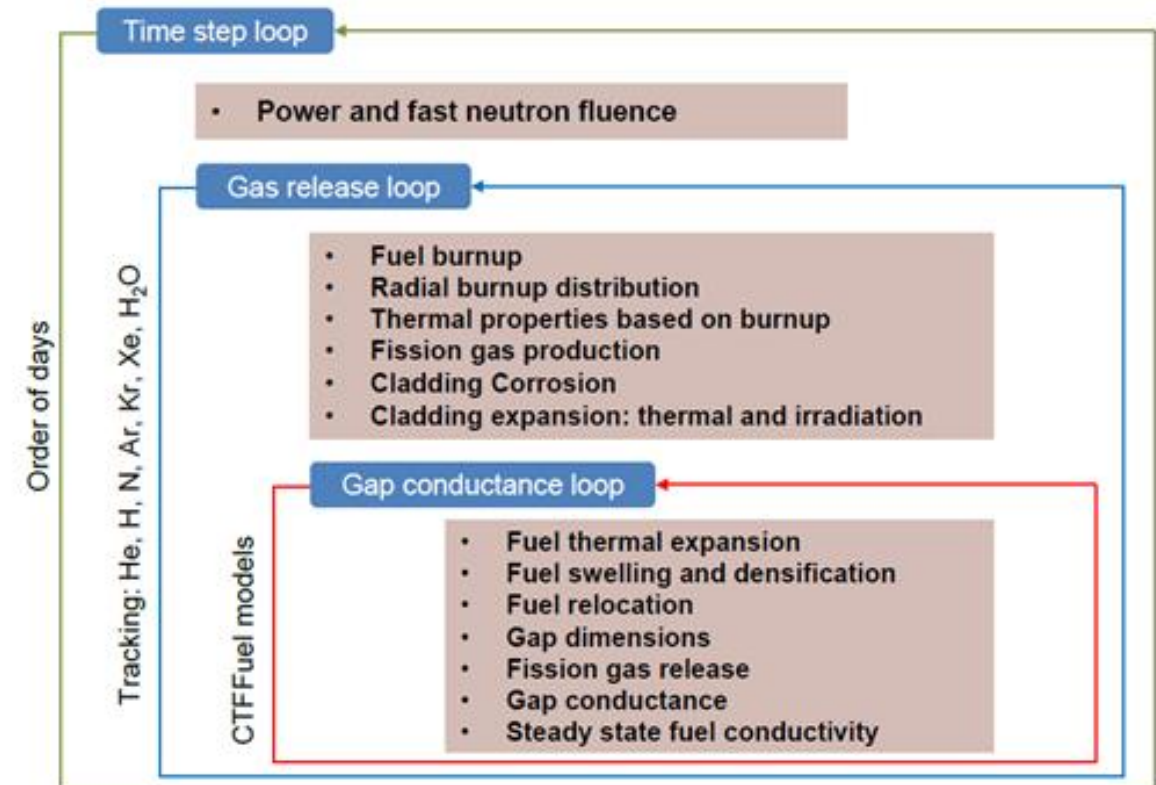
TOWARDS NOVEL CORE TH M&S

Improved Physics: Fuel behavior

- CTFFuel
 - The goal is developing of new fuel performance models for analyzing the fuel rod behavior to obtain accurate temperature profiles over the fuel rod mesh.



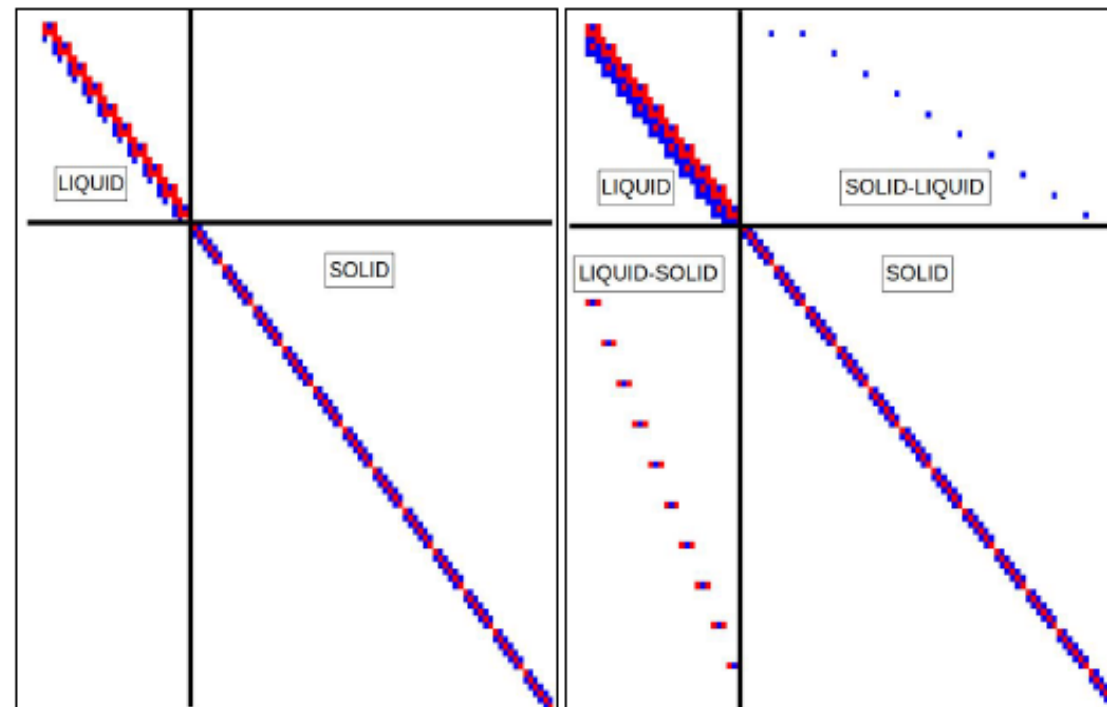
- Quasi steady-state burnup dependent fuel performance module
 - Simplify burnup model
 - Inter-pellet Radial Power Distribution (inter-pellet burnup)
 - Fission Gas Production and Release



TOWARDS NOVEL CORE TH M&S

Improved Numerics

- CTF-Residual (CTF-R)
 - The goal is to develop CTF version with numerically built Jacobian matrix that will allow to:
 - Add/remove or change conservation equations;
 - Can change the implicitness of the Jacobian;
 - Construction of the pressure or Jacobian matrix in a variety of ways;
 - Direct steady state solution.



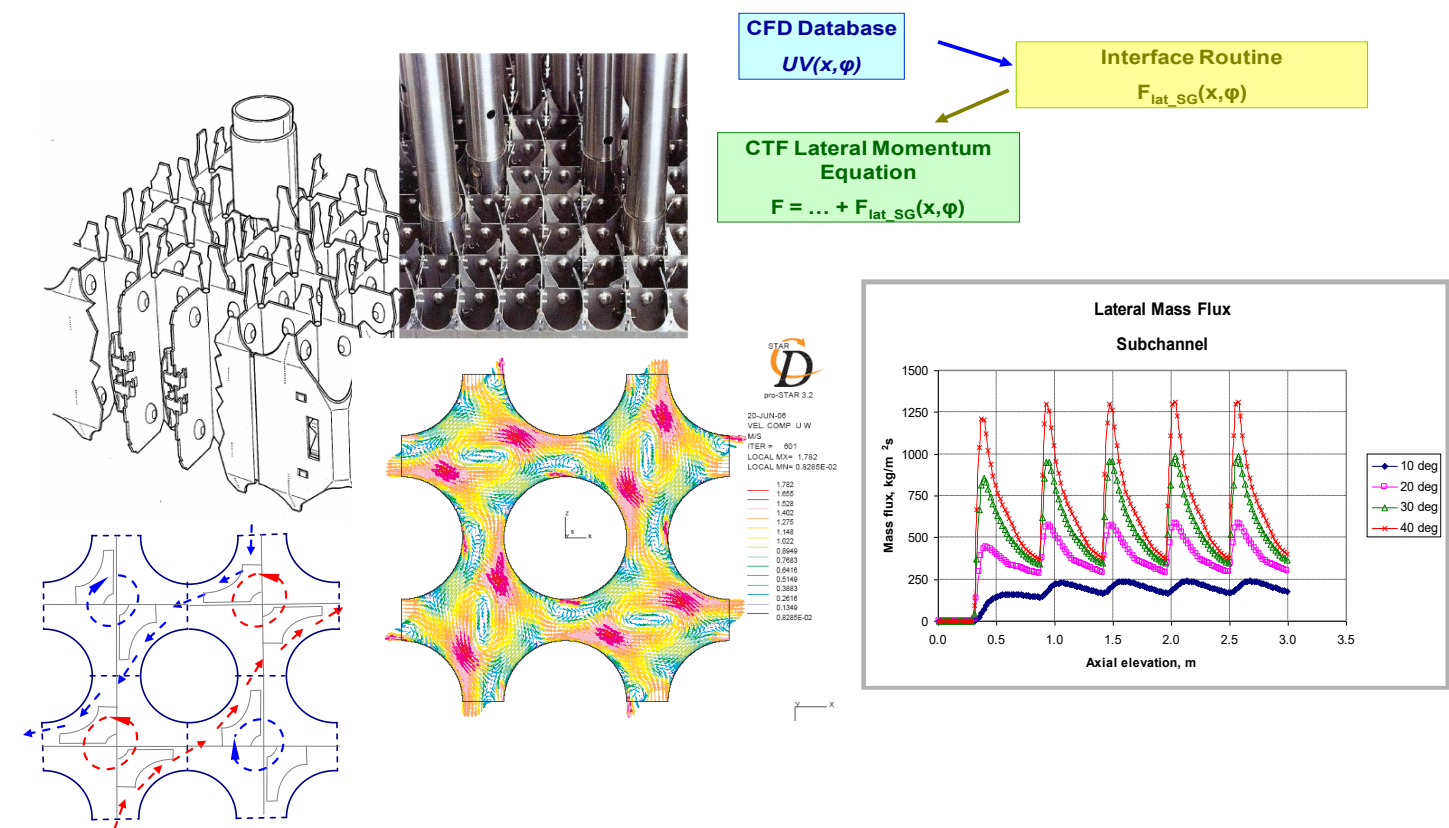
Explicitly (left) and Implicitly Coupled (right) Liquid Solid Jacobian Matrix

High-Fidelity Thermal-Hydraulics

The utilized thermal-hydraulic solvers in high-fidelity multi-physics simulations are:

- Improved (via off-line coupling with CFD) sub-channel codes (for multi-channel analysis).
- CFD in open media (detailed calculation within one sub-channel or within a plenum).

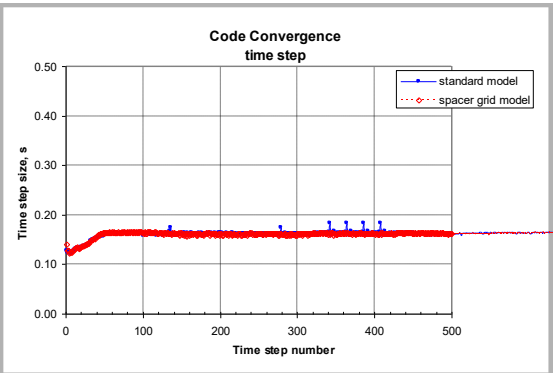
High-to-Low in TH: Grid-directed cross-flow model



Validation - SIEMENS Test Section 53 (TDS53) Mixing Tests

Run Number	Original COBRA-TF		New Model	
	Mean Value	Standard Deviation	Mean Value	Standard Deviation
8	8.6	1.4	1.5	0.8
9	7.5	1.7	1.3	0.6
10	8.1	1.6	1.3	1.0
11	4.7	1.0	0.7	0.3
12	4.7	1.2	0.8	0.3
13	4.4	0.9	0.6	0.5
14	2.9	0.9	0.7	0.4
15	2.6	0.9	0.6	0.3
16	5.0	2.0	1.8	1.2
17	5.4	1.9	1.4	1.0
18	6.0	1.9	1.3	1.0
19	8.0	1.6	1.6	1.2
20	7.2	3.0	2.0	2.0
21	7.2	2.9	2.0	1.8
22	8.2	2.7	1.9	1.4
23	7.6	2.5	1.8	1.2
24	7.2	2.8	2.1	1.3
25	8.5	3.4	2.3	2.1
26	9.2	4.3	3.1	2.9
27	10.2	2.5	2.4	1.9
28	9.8	2.3	3.7	2.1
average	6.8		1.6	

$$\overline{\Delta T}^{abs} = \frac{\sum_{n=1}^N |T_{calc} - T_{exp}|_n}{N}$$



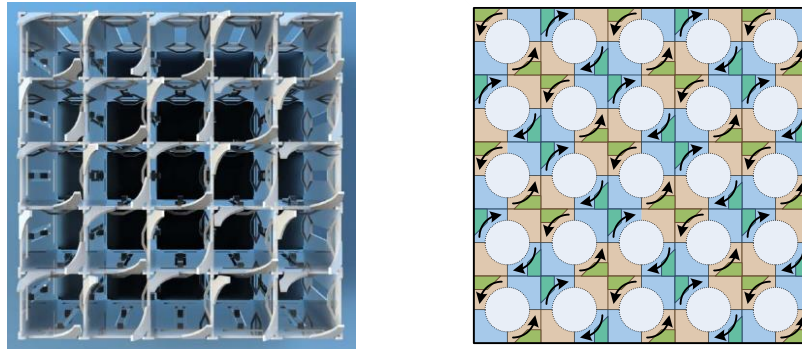
Measurement uncertainties: between 0.61 and 1.0 K for the absolute temperature in the range of 205 °C to 343 °C

Parameter	Range
Exit pressure, bar	159.2 ÷ 160.6
Fluid inlet temperature, °C	203.6 ÷ 282.1
Average mass flux, kg/m ² -s	3546.4 ÷ 3655.7
Bundle power, kW	1162.9 ÷ 4620.9

The CFD-informed model for grid-directed cross-flow (1) represents the advective nature of the directed standard crossflow; (2) is easy to implement; (3) is not computationally expensive.

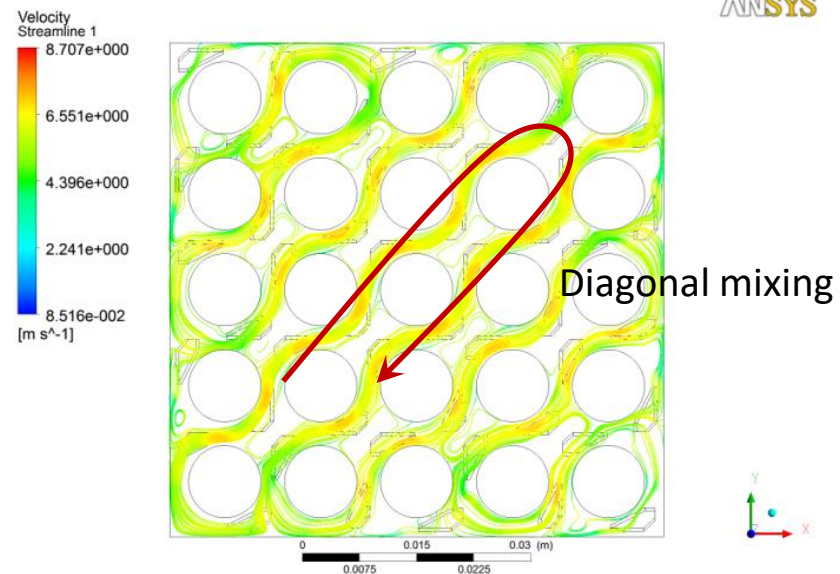
HIGH-FIDELITY THERMAL-HYDRAULICS

Mixing vane pattern



Local velocity(single grid)

At 40 mm downstream of MV



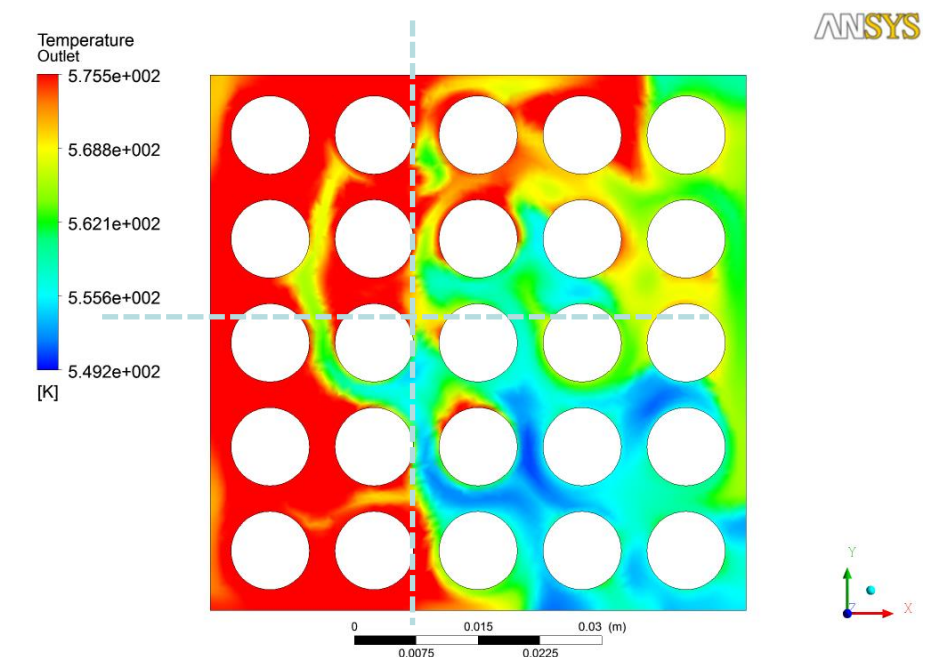
Temperature distribution

CFX 12.1

Standard k-e model

Full length with 7 MV

Temperature distribution at bundle exit

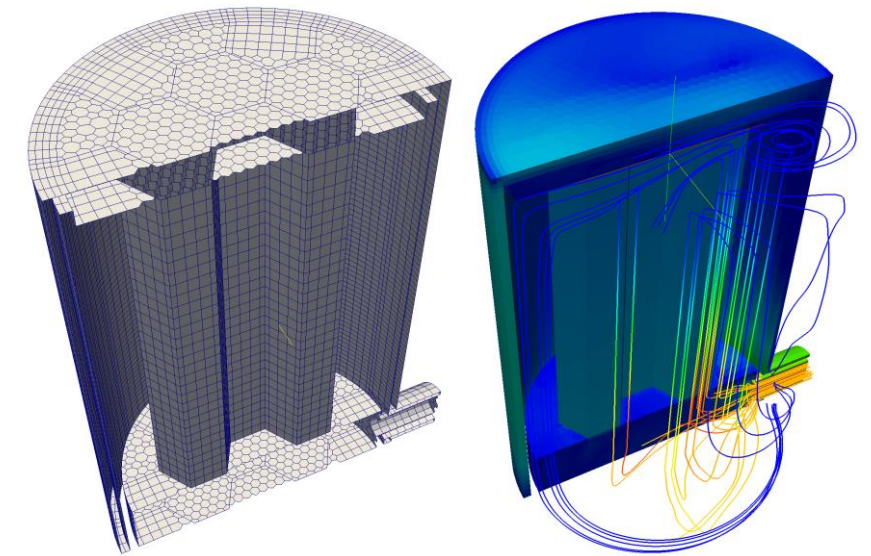
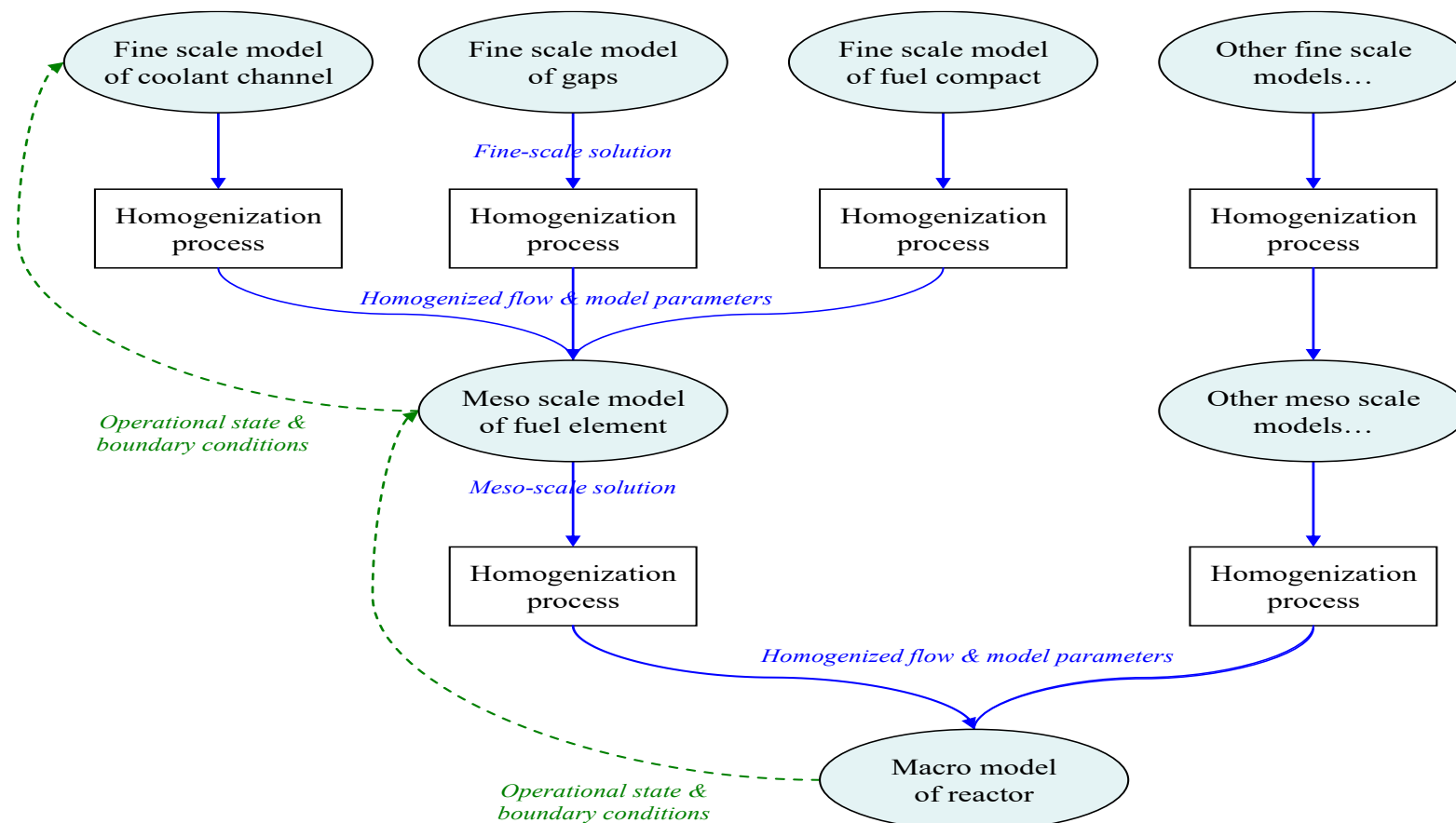


HIGH-FIDELITY THERMAL-HYDRAULICS

Coupled system-CFD models:

- Multi-scale hierarchical CFD approach using hybrid coarse/fine-mesh solvers and mathematically consistent homogenization techniques on multi-physics platforms such as OpenFOAM

Flow-chart of multi-scale hierarchical CFD approach



Coarse mesh CFD modeling and solution for a small representative prismatic HTR

CFD-INFORMED MODELS IN CTF

Spacer Grid Enhanced Heat Transfer

- Heat transfer from the rod surfaces to the coolant near a spacer grid is affected due to changes in the local flow
 - Asymmetries in spacer grid designs impact the flow differently at the sub-channel level
- Common issues with the heat transfer enhancement correlations (such as Yao-Hochreiter-Leech correlation)
 - Requires detailed knowledge of spacer grid dimensions, which may portend geometry assumptions for proprietary designs
 - Difficult to apply at a subchannel level of refinement → No method for specifying different values at the rod surface level
 - Applied in CTF starting at the leading-edge location and not necessarily where the peak HTC value exists
 - No upstream HTC effects possible

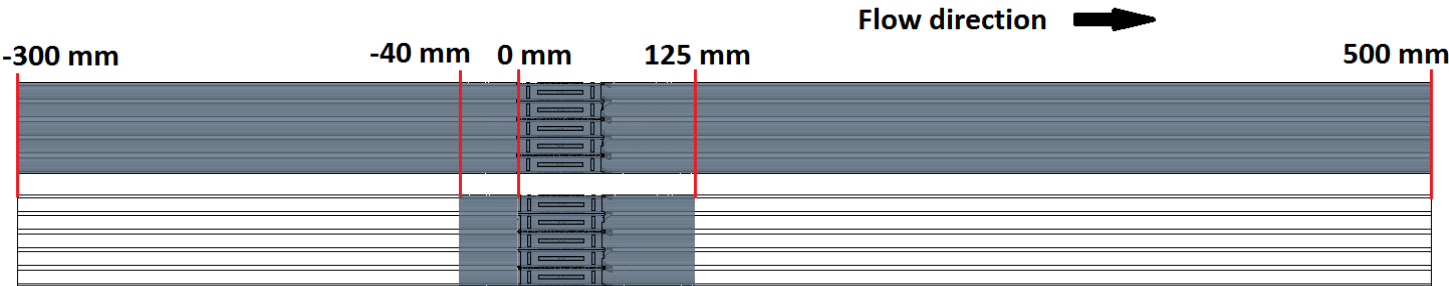
CFD-INFORMED MODELS IN CTF

Spacer Grid Pressure Losses

- CTF accounts for pressure losses due to spacer grids using a form loss model:

$$\Delta P_{form} = \frac{1}{2} \zeta_{grid} \rho u^2$$

- CFD simulation details:
 - 5x5 rod bundle layout is considered for the PSBT spacer grids
 - Spacer grid leading edge placed at 0.0 mm axially
 - 15.5 MPa, 300°C inlet, 1.276 kg/s flow rate
- CFD results of interest:
 - STAR-CCM+ predicted pressures averaged at axial planes



- CFD results for the PSBT 5x5 rod bundle simulations
(Performed at 5.0 m/s axial flow velocity)

PSBT spacer grid	MVG	NMV	SSG
Total pressure drop	27.3 kPa	22.0 kPa	20.4 kPa
Frictional pressure drop	11.7 kPa	11.7 kPa	11.7 kPa
Form pressure drop	15.6 kPa	10.3 kPa	8.7 kPa

PSBT Grid	CFD-calculated loss coefficient
MVG	1.74
NMV	1.15
SSG	0.97

$$\Delta P_{grid,CFD} = \Delta P_{total,CFD} - \Delta P_{fric,CFD}$$

$$\zeta_{grid,CFD} = 2 \frac{\Delta P_{grid,CFD}}{\rho u^2}$$

ζ_{grid_MVG} : 1.0	ζ_{grid_MVG} : 1.74
ζ_{grid_NMV} : 0.7	ζ_{grid_NMV} : 1.15
ζ_{grid_SSG} : 0.4	ζ_{grid_SSG} : 0.97

PSBT spacer grid	MVG	NMV	SSG
CTF (PSBT-recommended loss coefficient)	9.1 kPa	6.4 kPa	3.6 kPa
CTF (CFD-determined loss coefficient)	15.8 kPa	10.4 kPa	8.8 kPa
STAR-CCM+	15.6 kPa	10.3 kPa	8.7 kPa

TOWARDS NOVEL CORE TH M&S - SUMMARY

- The developments in thermal-hydraulic sub-channel modeling for whole core multi-physics simulations are towards:
 - Acceleration and parallelization for improving efficiency of sub-channel core simulations especially on pin-resolved level;
 - Development and implementation of models for improving accuracy of predictions of feedback parameters for neutronics core calculations:
 - For LWRs the predictions of the following feedback parameters are important: fuel (Doppler) temperature, coolant (moderator) temperature and density (void fraction), and boron concentration;
 - Traditional mechanistic modeling approach could be efficiently supplemented/replaced by high-to-low model information approach and physics-informed data-driven modeling;
 - The development and implementation of residual formulations of sub-channel codes will help to address both improving the efficiency and accuracy of sub-channel calculations as well as improving the coupling with neutronics codes towards fully implicit coupling.

HIGH-FIDELITY FUEL PERFORMANCE

HIGH-FIDELITY FUEL MODELING

High Fidelity (e.g. BISON, OFFBEAT, ALCYONE):

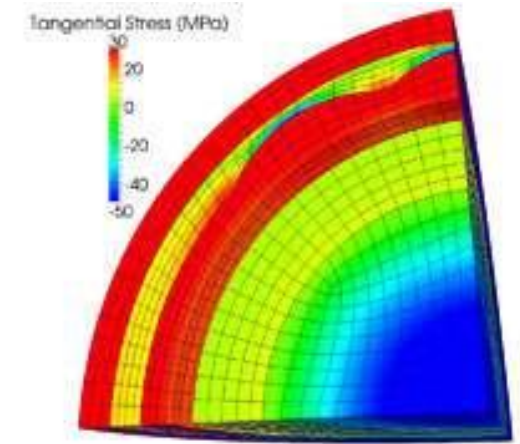
- Finite elements or Finite volumes 3D/2D or 1.5D.
- Full coupling of thermal, mechanical and physico-chemical aspects.
- Empirical models for most important phenomena.
- Computationally very expensive and mainly used in R&D.

Standard Fidelity (e.g. FAST/FRAPCON/FRAPTRAN, TRANSURANUS):

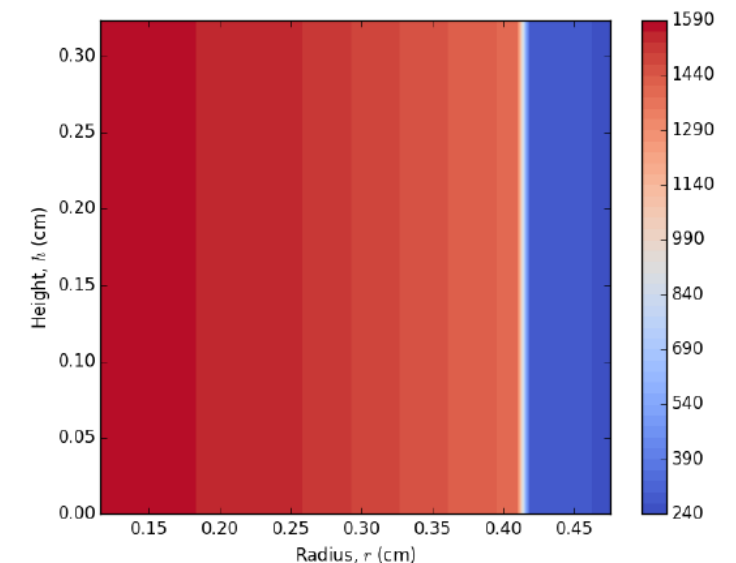
- Finite differences 1.5D.
- Full coupling of thermal, mechanical and physico-chemical aspects.
- Empirical models for most important phenomena.
- Computationally very efficient and used in both R&D and licensing.

Low Fidelity (e.g. CTFFuel, FINIX):

- Finite differences 1D or 1.5D.
- Solve only heat conduction in fuel and cladding by imposing the gap heat transfer. Few thermal phenomena are considered.
- Computationally very cheap and usually consist thermal solvers of Thermal-Hydraulics codes.



*<https://bison.inl.gov/SitePages/Applications.aspx>

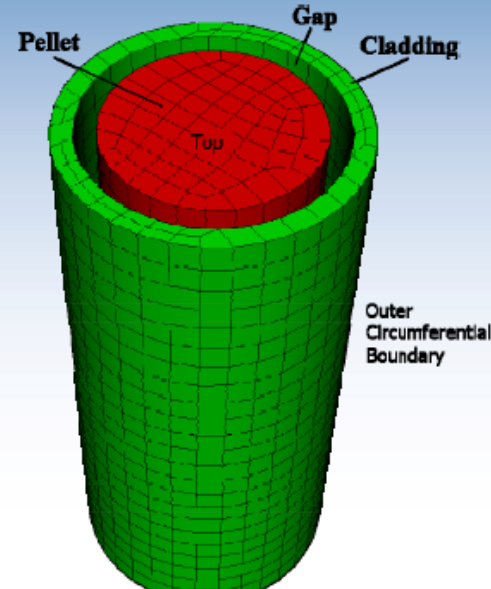


HIGH-FIDELITY FUEL MODELING

Fuel Performance Modeling

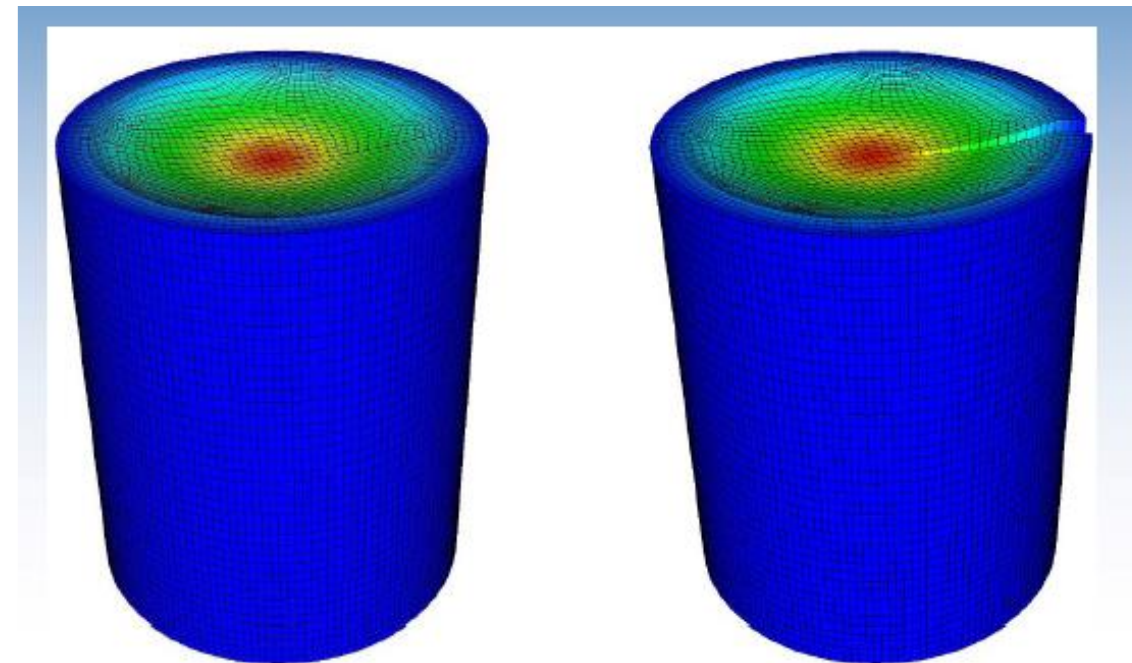
- Robust and efficient solvers
- 3D coupling to 2D (and 1D) in the same code
- Tight coupling between physics
 - contact and thermal expansion
 - fission gases in gap modify heat transfer
- Resolve small time scale effects accurately
 - implicit to take long time steps during high burnup phase
- Excellent serial and small parallel system performance
 - massively parallel not ignored

- Details of fuel pellet / cladding mechanical interaction
- Fuel thermomechanics
- Fission gas formation / migration
- Restructuring, constituent migration
- Damage – radiation, cracking, corrosion



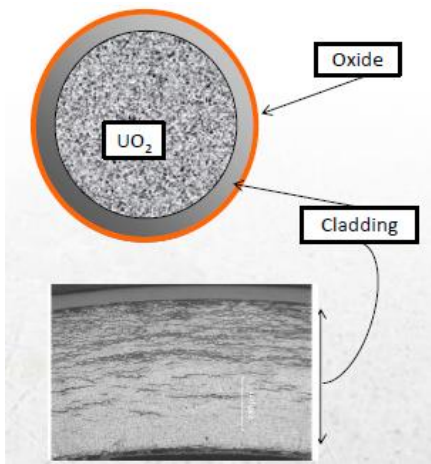
Fuels Performance Code Architecture

- Requirements:
 - 3D (and 2D)
 - Massively parallel
 - Fully coupled and implicit
 - Advanced solution strategies (adaptivity, etc.)
 - Portable
 - Flexible physics interface
 - Flexible materials database

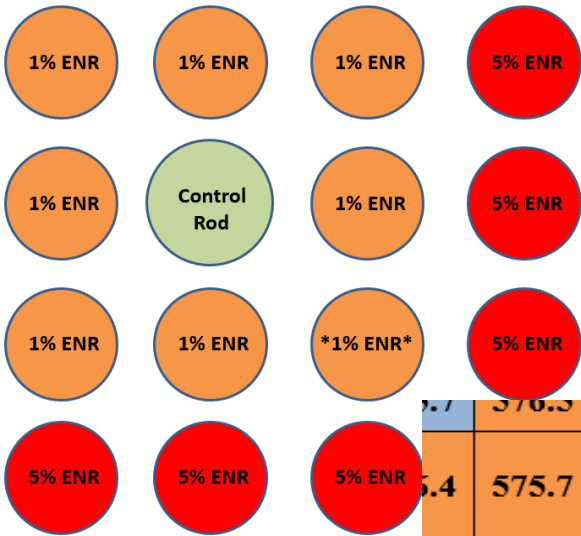


HIGH-FIDELITY FUEL MODELING

Example: Prediction of Anisotropic Azimuthal Power and Temperature Distribution as a Driving Force for Hydrogen Redistribution

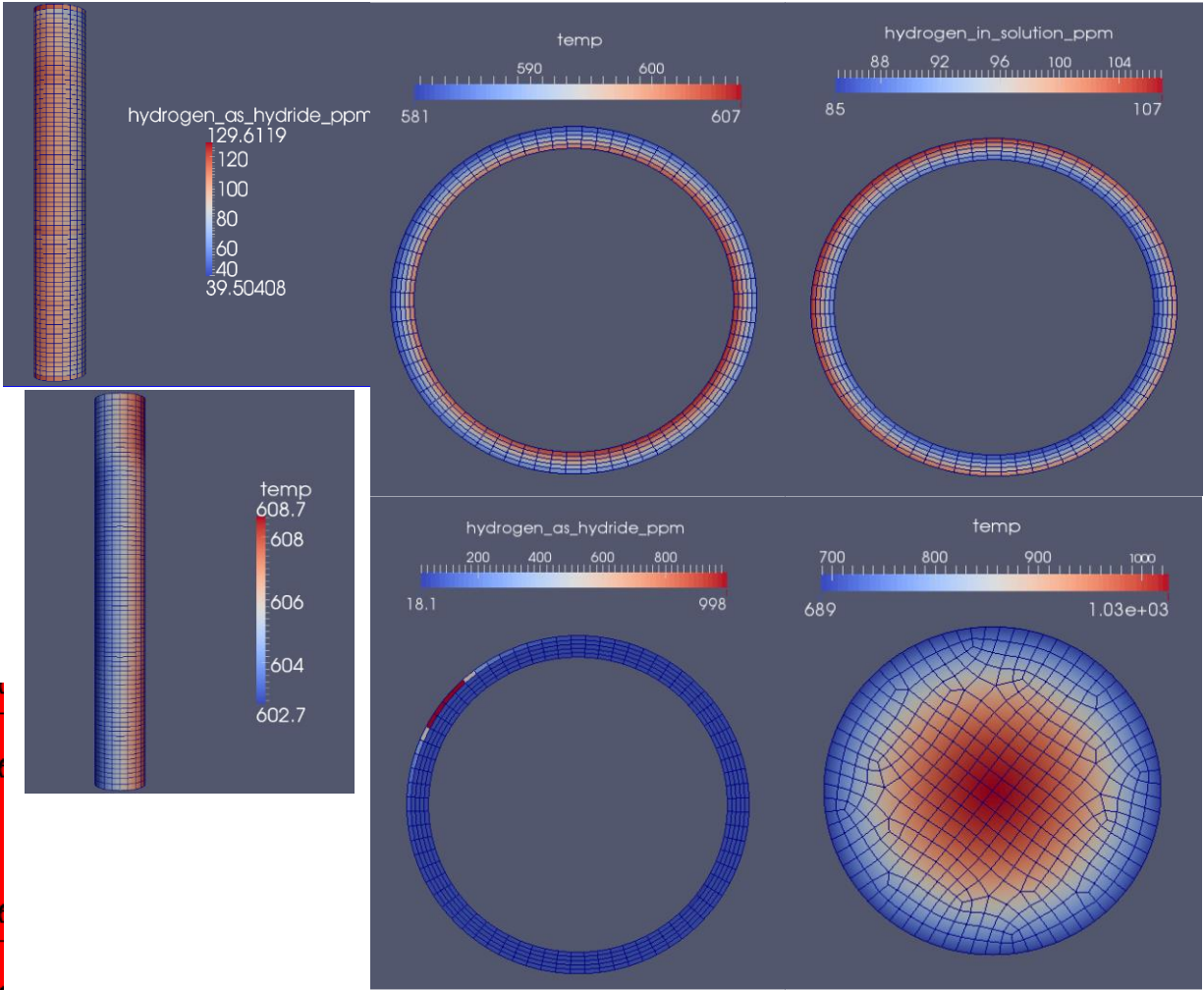


4x4 Sub-Assembly with Control Rod layout at 0 MWd/kgU Burnup



575.7	576.5	583.1
575.4	575.7	582.6
1% Enr.		
582.2	582.5	585.4
584.4	587.2	

Azimuthal Outer Cladding Temperatures (K)



Top Left: Cladding temperature (K) Top Right: Cladding hydrogen dist.
Bottom left: Cladding hydride dist. Bottom Right: Fuel temperature (K)

Source: DOE NEUP project 11-2987

HIGH-FIDELITY FUEL MODELING

High burnup fuel poses new challenges in advanced modeling and simulation:

- Cladding integrity
- Corrosion and CRUD
- Hydriding
- Creep
- Fuel cladding mechanical interactions
- High burnup structure
- Fuel fragmentation, relocation, and dispersal (FFRD)
- Grid-to-rod fretting (GTRF)

Next Class: *Multi-Scale Single Physics Modeling Using Hi2Lo Information Schemes and Implementation within Multi-Physics Calculations*